

**ENGINEERING MATHEMATICS - I**

**UNIT - 3 : DIFFERENTIAL CALCULUS - II**

**ONE SHOT LECTURE**

**CRASH COURSE**



**By Gulshan sir**



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- Expansion of functions
  - (1) Taylor's Series for one Variables
  - (2) Maclaurin's Series for one Variables
  - (3) Taylor's Series for two Variables
  - (4) Maclaurin's Series for two Variables
- Maxima and Minima for the functions of two variables
- Maxima and Minima for the functions of three variables : **Lagrange's method of multipliers**
- Approximation of errors
- Jacobians
  - (1) Basic of Jacobians
  - (2) Properties of Jacobians
  - (3) Jacobians of Implicit function



## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 1

##### Today's target

- Maclaurin's Theorems / Maclaurin's series  
(For the function of one Variable)
- PYQs
- DPP



By Gulshan sir



## UNIT- 3 : Differential Calculus-II

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- Expansion of functions by Taylor's and Maclaurin's theorems for functions of one and two Variables
- Maxima and Minima of functions of several variables
- Lagrange's method of multipliers
- Jacobians
- Approximation of errors

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## Expansion of functions

Expansion of function of one variable

(i) By Maclaurin's theorem

(ii) By Taylor's theorem

Expansion of function of two variable

(i) By Taylor's theorem

(ii) By Maclaurin's theorem

Gateway Classes



$$(1 + x)^n = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots \dots \dots \text{Where } |x| < 1$$

(i)  $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \dots \dots \dots$

(ii)  $(1 + x)^{-1} = 1 - x + x^2 - x^3 + \dots \dots \dots$

(iii)  $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots \dots \dots$

(iv)  $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots \dots \dots$

### $n^{\text{th}}$ Derivative

(i)  $\frac{d^n}{dx^n} e^{ax} \sin(bx + c) = (a^2 + b^2)^{n/2} e^{ax} \sin(bx + c + n \tan^{-1} \frac{b}{a})$

(i)  $\frac{d^n}{dx^n} e^{ax} \cos(bx + c) = (a^2 + b^2)^{n/2} e^{ax} \cos(bx + c + n \tan^{-1} \frac{b}{a})$

## Maclaurin's theorem (Maclaurin's Series)

Let  $f(x)$  be a function of  $x$  which can be expanded in ascending power of  $x$  and  $f(x)$  can be differentiable any number of times.

Then,

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \frac{x^5}{5!} f^v(0) + \dots$$

### Working Rule

**Step – 1 :** Put  $f(x) =$  given function

**Step – 2 :** Find  $f'(x)$ ,  $f''(x)$ ,  $f'''(x)$ ,  $f^{iv}(x)$ ,  $f^v(x)$ ,  $f^{vi}(x)$  ... ..

**Step – 3 :** Put  $x = 0$  and find  $f(0)$ ,  $f'(0)$ ,  $f''(0)$ ,  $f'''(0)$ ,  $f^{iv}(0)$ ,  $f^v(0)$ ,  $f^{vi}(0)$  .....

**Step – 4 :** Put  $f(0)$ ,  $f'(0)$ ,  $f''(0)$ ,  $f'''(0)$ ,  $f^{iv}(0)$ ,  $f^v(0)$ ,  $f^{vi}(0)$  .... in Maclaurin's Series



# Q.1 Expand $e^x$ in powers of $x$ by Maclaurin's Theorem

$$\text{Let } f(x) = e^x \quad \left| \quad f(0) = 1\right.$$

$$f'(x) = e^x \quad \left| \quad f'(0) = 1\right.$$

$$f''(x) = e^x \quad \left| \quad f''(0) = 1\right.$$

$$f'''(x) = e^x \quad \left| \quad f'''(0) = 1\right.$$

By Maclaurin theorem

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

Q.2 Expand  $\sin x$  in powers of  $x$  by Maclaurin's Theorem

Let  $f(x) = \sin x$

$f'(x) = \cos x$

$f''(x) = -\sin x$

$f'''(x) = -\cos x$

$f^{(4)}(x) = \sin x$

$f^{(5)}(x) = \cos x$

$f(0) = 0$

$f'(0) = 1$  ✓

$f''(0) = 0$

$f'''(0) = -1$  ✓

$f^{(4)}(0) = 0$

$f^{(5)}(0) = 1$  ✓

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0) + \frac{x^5}{5!} f^{(5)}(0) + \dots$$

$$\sin x = 0 + x + 0 - \frac{x^3}{3!} + 0 + \frac{x^5}{5!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

By Maclaurin's Theorem



Q.3 Expand  $\tan x$  by Maclaurin's Theorem as far as  $x^5$  and hence find the value of  $\tan 46^\circ 30'$  up to four decimal places.

$$\text{Let } f(x) = \tan x \Rightarrow \boxed{f(0) = 0}$$

$$f'(x) = \sec^2 x$$

$$f'(x) = 1 + \tan^2 x \Rightarrow \boxed{f'(0) = 1}$$

$$f''(x) = 2 \tan x \sec^2 x$$

$$f''(x) = 2 \tan x (1 + \tan^2 x)$$

$$f''(x) = 2 \tan x + 2 \tan^3 x \Rightarrow \boxed{f''(0) = 0}$$

$$f'''(x) = 2 \sec^2 x + 6 \tan^2 x \sec^2 x$$

$$f'''(x) = 2(1 + \tan^2 x) + 6 \tan^2 x (1 + \tan^2 x)$$

$$f'''(x) = 2 + 2 \tan^2 x + 6 \tan^2 x + 6 \tan^4 x \Rightarrow \boxed{f'''(0) = 2}$$

$$f^{(iv)}(x) = 4 \tan x \sec^2 x + 12 \tan x \sec^2 x + 24 \tan^3 x \sec^2 x$$

$$f^{(iv)}(x) = 4 \tan x (1 + \tan^2 x) + 12 \tan x (1 + \tan^2 x) + 24 \tan^3 x (1 + \tan^2 x)$$

$$f^{iv}(x) = 4 \tan x + 4 \tan^3 x + 12 \tan x + 12 \tan^3 x + 24 \tan^3 x + 24 \tan^5 x$$

$$f^{iv}(x) = 16 \tan x + 40 \tan^3 x + 24 \tan^5 x \Rightarrow f^{iv}(0) = 0$$

$$f^v(x) = 16 \sec^2 x + 120 \tan^2 x \sec^2 x + 120 \tan^4 x \sec^2 x \Rightarrow f^v(0) = 16$$

By Maclaurin's theorem

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \frac{x^5}{5!} f^v(0) + \dots$$

$$\tan x = 0 + x + 0 + \frac{2x^3}{3!} + 0 + \frac{16x^5}{5!} + \dots$$

$$\tan x = x + \frac{x^3}{3} + \frac{2}{15} x^5 + \dots \quad \text{--- (1)}$$



Note  $1^\circ = 60'$

$$1' = 60''$$

$$\Rightarrow 60' = 1^\circ$$

$$30' = \left(\frac{30}{60}\right)^\circ = \left(\frac{1}{2}\right)^\circ$$

$$\alpha = 46^\circ 30'$$

$$\alpha = \left(46\frac{1}{2}\right)^\circ$$

$$\alpha = \left(\frac{93}{2}\right)^\circ$$

$$\alpha = \frac{93}{2} \times \frac{\pi}{180}$$

$$\alpha = 0.812$$

Put  $\alpha$  in ①

$$\tan 46^\circ 30' = 0.812 + \frac{(0.812)^3}{3} + \frac{2(0.812)^5}{15}$$

$$\tan 46^\circ 30' = 1.0374$$

**Q.4** Expand  $\log(1+x)$  in powers of  $x$ . Then find series for  $\log_e \left( \frac{1+x}{1-x} \right)$  and hence determine the value of  $\log_e \left( \frac{11}{9} \right)$  upto five places of decimal.

$$\text{Let } f(x) = \log(1+x) \Rightarrow f(0) = 0$$

$$f'(x) = \frac{1}{1+x}$$

$$f'(x) = (1+x)^{-1} \Rightarrow f'(0) = 1$$

$$f''(x) = -1(1+x)^{-2} \Rightarrow f''(0) = -1$$

$$f'''(x) = (-1)(-2)(1+x)^{-3} \Rightarrow f'''(0) = 2$$

$$f^{(4)}(x) = (-1)(-2)(-3)(1+x)^{-4} \Rightarrow f^{(4)}(0) = -6$$

By Maclaurin's Theorem

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0) + \dots$$

$$\log(1+x) = 0 + x - \frac{x^2}{2!} + \frac{2x^3}{3!} - \frac{6x^4}{4!} + \dots$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

Similarly

$$\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \dots$$



$$\log\left(\frac{1+n}{1-n}\right) = \log(1+n) - \log(1-n)$$

$$= n - \frac{n^2}{2} + \frac{n^3}{3} - \frac{n^4}{4} + \dots + n + \frac{n^2}{2} + \frac{n^3}{3} + \frac{n^4}{4} + \dots$$

$$\log\left(\frac{1+n}{1-n}\right) = 2\left(n + \frac{n^3}{3} + \frac{n^5}{5} + \dots\right)$$

Put  $n = \frac{1}{10}$

$$\log\left(\frac{11}{9}\right) = 2\left[\frac{1}{10} + \frac{1}{3}\left(\frac{1}{10}\right)^3 + \frac{1}{5}\left(\frac{1}{10}\right)^5 + \dots\right]$$

$$\log\left(\frac{11}{9}\right) = 0.20067 \text{ approx}$$

Q.5 Expand  $\tan^{-1}x$  up to four terms in power of  $x$

$$\text{Let } f(x) = \tan^{-1}x \Rightarrow f(0) = 0$$

$$f'(x) = \frac{1}{1+x^2}$$

$$f'(x) = (1+x^2)^{-1}$$

Apply Binomial Theorem

$$f'(x) = 1 - x^2 + x^4 - x^6 + \dots \quad \boxed{f'(0) = 1}$$

$$f''(x) = 0 - 2x + 4x^3 - 6x^5 + \dots \quad \boxed{f''(0) = 0}$$

$$f'''(x) = -2 + 12x^2 - 30x^4 + \dots \quad \boxed{f'''(0) = -2}$$

$$f^{(iv)}(x) = 24x - 120x^3 + \dots \Rightarrow \boxed{f^{(iv)}(0) = 0}$$

$$f^{(v)}(x) = 24 - 360x^2 + \dots \Rightarrow \boxed{f^{(v)}(0) = 24}$$

$$f^{(vi)}(x) = -720x + \dots \Rightarrow \boxed{f^{(vi)}(0) = 0}$$

$$f^{(vii)}(x) = -720 \Rightarrow \boxed{f^{(vii)}(0) = -720}$$

By Maclaurin's theorem



$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0) + \frac{x^5}{5!} f^{(5)}(0) + \frac{x^6}{6!} f^{(6)}(0) + \frac{x^7}{7!} f^{(7)}(0) + \dots$$

$$\tan^{-1}x = 0 + x + 0 - \frac{2x^3}{3!} + 0 + \frac{24}{5!}x^5 + 0 - \frac{720x^7}{7!} + \dots$$

$$\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

Q.6 Expand  $e^{2x} \sin x$  in ascending powers of  $x$  up to  $x^5$

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$$\text{Let } f(x) = e^{2x} \sin x \Rightarrow f(0) = 0$$

$$f'(x) = e^{2x} \cos x + 2e^{2x} \sin x \Rightarrow f'(0) = 1$$

$$f'(x) = e^{2x} (\cos x + 2 \sin x)$$

$$f''(x) = e^{2x} (-\sin x) + 2e^{2x} \cos x + 2f'(x) \Rightarrow f''(0) = 2 + 2 = 4$$

$$f''(x) = -f(x) + 2e^{2x} \cos x + 2f'(x)$$

$$f'''(x) = -f'(x) + 2e^{2x} (-\sin x) + 4e^{2x} \cos x + 2f''(x) \Rightarrow f'''(0) = -1 + 4 + 8 = 11$$

$$f'''(x) = -f'(x) - 2f(x) + 4e^{2x} \cos x + 2f''(x)$$



$$f^{iv}(n) = -f''(n) - 2f'(n) + 4e^{2n}(-\sin n) + 8e^{2n}\cos n + 2f'''(n) \Rightarrow f^{iv}(0) = -4 - 2 + 8 + 2 \times 11$$

$$f^{iv}(0) = 24$$

$$f^{iv}(n) = -f''(n) - 2f'(n) - 4f(n) + 8e^{2n}\cos n + 2f'''(n)$$

$$f^{iv}(n) = -f'''(n) - 2f''(n) - 4f'(n) + 8e^{2n}(-\sin n) + 16e^{2n}\cos n + 2f^{iv}(n)$$

$$\Rightarrow f^{iv}(0) = -11 - 8 - 4 + 0 + 16 + 48$$

$$f^{iv}(0) = 41$$

By Maclaurin's series

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0) + \frac{x^5}{5!} f^{(5)}(0) + \dots$$

$$e^{2x} \sin x = 0 + x + \frac{4x^2}{2!} + \frac{11x^3}{3!} + \frac{24x^4}{4!} + \frac{41x^5}{5!} + \dots$$



UNIT -3 : DPP - 1

Topic : Maclaurin's Theorems/ Maclaurin's series

Q.1 Expand  $\sec x$  in powers of  $x$  by Maclaurin's Theorem

Q.2 Expand  $\sin^{-1} x$  up to four terms in powers of  $x$

Q.3 Expand by Maclaurin's Theorem  $\frac{e^x}{e^x+1}$ , as far as  $x^3$

Q.4 Prove by Maclaurin's Theorem that  $e^{\sin x} = 1 + x + \frac{x^2}{1.2} - \frac{3 \cdot x^4}{1.2.3.4} + \dots$

Q.5 Show that  $\log \frac{\tan x}{x} = \frac{x^2}{3} + \frac{7}{90}x^4 + \dots$

Q.6 Show that  $\tan^{-1} \frac{2x}{1-x^2} = \sin^{-1} \left( \frac{2x}{1+x^2} \right) = 2 \left( x - \frac{x^3}{3} + \frac{x^5}{5} - \dots \right)$

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Q.7 Expand  $e^x \sin x$  in ascending powers of  $x$  up to  $x^5$

## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 2

##### Today's target

- Taylor's Theorems / Taylor's series  
(For the function of one Variable)
- PYQs
- DPP



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## Taylor's Theorem (Taylor's Series)

A differentiable function  $f(x + h)$  can be expanded in **power of  $h$  or  $x$**  by Taylor's Theorem

$$(i) f(x + h) = f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots$$

$$(ii) f(x + h) = f(h) + x f'(h) + \frac{x^2}{2!} f''(h) + \frac{x^3}{3!} f'''(h) + \dots$$

V. imp

Expansion in power of  $(x - a)$

OR

Expansion About  $x = a$

$$f(\underbrace{x - a}_x + \underbrace{a}_h) = f(a) + (x - a) f'(a) + \frac{(x - a)^2}{2!} f''(a) + \frac{(x - a)^3}{3!} f'''(a) + \dots$$

Q.1 Use Taylor's Theorem to express the polynomial  $2x^3 + 7x^2 + x - 6$  in powers of  $(x - 2)$

$$\text{Let } f(x) = 2x^3 + 7x^2 + x - 6$$

By Taylor's Theorem

$$f\left(\underbrace{x-2}_x + \underbrace{2}_h\right) = f(2) + (x-2)f'(2) + \frac{(x-2)^2}{2!}f''(2) + \frac{(x-2)^3}{3!}f'''(2) + \dots \quad \text{--- ①}$$

$$f(x) = 2x^3 + 7x^2 + x - 6 \Rightarrow f(2) = 16 + 28 + 2 - 6 = 40$$

$$f'(x) = 6x^2 + 14x + 1 \Rightarrow f'(2) = 24 + 28 + 1 = 53$$

$$f''(x) = 12x + 14 \Rightarrow f''(2) = 24 + 14 = 38$$

$$f'''(x) = 12 \Rightarrow f'''(2) = 12$$



Put these values in ①

$$f(n) = 40 + 53(n-2) + \frac{38}{2}(n-2)^2 + \frac{12}{3 \times 2 \times 1}(n-2)^3$$

$$2n^3 + 7n^2 + n - 6 = 40 + 53(n-2) + 19(n-2)^2 + 2(n-2)^3$$

Q.2 Expand  $\sin x$  in ascending powers of  $(x - \frac{\pi}{2})$ .

Let  $f(x) = \sin x$

By Taylor's Theorem

$$f\left(\underbrace{x - \frac{\pi}{2}}_x + \underbrace{\frac{\pi}{2}}_h\right) = f\left(\frac{\pi}{2}\right) + (x - \frac{\pi}{2})f'\left(\frac{\pi}{2}\right) + \frac{(x - \frac{\pi}{2})^2}{2!}f''\left(\frac{\pi}{2}\right) + \frac{(x - \frac{\pi}{2})^3}{3!}f'''\left(\frac{\pi}{2}\right) + \dots$$

$$f(x) = \sin x \Rightarrow f\left(\frac{\pi}{2}\right) = 1$$

$$f'(x) = \cos x \Rightarrow f'\left(\frac{\pi}{2}\right) = 0$$

$$f''(x) = -\sin x \Rightarrow f''\left(\frac{\pi}{2}\right) = -1$$

$$f'''(x) = -\cos x \Rightarrow f'''\left(\frac{\pi}{2}\right) = 0$$



$$f^{(r)}(x) = \sin x \Rightarrow f^{(4)}\left(\frac{\pi}{2}\right) = 1$$

Put these values in ①

$$f(x) = 1 + 0 - \frac{\left(x - \frac{\pi}{2}\right)^2}{2!} + 0 + \frac{\left(x - \frac{\pi}{2}\right)^4}{4!} + \dots$$

$$\sin x = 1 - \frac{\left(x - \frac{\pi}{2}\right)^2}{2!} + \frac{\left(x - \frac{\pi}{2}\right)^4}{4!} + \dots$$

Q.3 Expand  $\log x$  in powers of  $(x - 1)$  by Taylor's theorem and hence find the value of  $\log_e(1.1)$

$$\text{Let } f(x) = \log x$$

$$\frac{d}{dx} (x^n) = n x^{n-1}$$

By Taylor's Theorem

$$f\left(\underbrace{x-1}_{h} + 1\right) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1) + \dots \quad \text{--- (1)}$$

$$f(x) = \log x \Rightarrow f(1) = 0$$

$$f'(x) = \frac{1}{x} \Rightarrow f'(1) = 1$$

$$f''(x) = -\frac{1}{x^2} \Rightarrow f''(1) = -1$$

$$f'''(x) = \frac{2}{x^3} \Rightarrow f'''(1) = 2$$



Put these values in ①

$$f(n) = 0 + (n-1) - \frac{(n-1)^2}{2!} + \frac{2(n-1)^3}{3!} + \dots$$

$$\log n = (n-1) - \frac{(n-1)^2}{2} + \frac{(n-1)^3}{3} + \dots$$

$$\log(1.1) = (1.1-1) - \frac{(1.1-1)^2}{2} + \frac{(1.1-1)^3}{3} + \dots$$

$$\log(1.1) = 0.0953$$



Let  $f(x) = \tan^{-1}x$

By Taylor's Theorem

$$f\left(\frac{x-1+h}{h}\right) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \dots \quad \text{--- (1)}$$

$$f(x) = \tan^{-1}x$$

$$\Rightarrow f(1) = \tan^{-1}(1) = \frac{\pi}{4}$$

$$f'(x) = \frac{1}{1+x^2}$$

$$\Rightarrow f'(1) = \frac{1}{1+1} = \frac{1}{2}$$

$$f''(x) = \frac{-1 \times 2x}{(1+x^2)^2}$$

$$\Rightarrow f''(1) = \frac{-2}{4} = -\frac{1}{2}$$



Put these values in ①

$$f(n) = \frac{\pi}{4} + \frac{n-1}{2} - \frac{(n-1)^2}{2 \times 2!} + \dots$$

$$\tan^{-1} n = \frac{\pi}{4} + \frac{(n-1)}{2} - \frac{(n-1)^2}{4} + \dots$$

Q.5 Expand  $\log(x + h)$  in powers of  $x$  up to four terms by Taylor's series

$$\text{Let } f(n+h) = \log(n+h)$$

$$\text{Put } n=0$$

$$f(h) = \log h$$

By Taylor's theorem

$$f(n+h) = f(h) + n f'(h) + \frac{n^2}{2!} f''(h) + \frac{n^3}{3!} f'''(h) + \dots \quad \text{①}$$

$$f'(h) = \frac{1}{h}$$

$$f''(h) = -\frac{1}{h^2}$$

$$f'''(h) = \frac{2}{h^3}$$

Put these values in ①

$$\log(n+h) = \log h + \frac{n}{h} - \frac{n^2}{2h^2} + \frac{n^3}{3h^3} + \dots$$



Q.6 Expand  $\sin^{-1}(x+h)$  in powers  $h$

Let  $f(x+h) = \sin^{-1}(x+h)$

Put  $h = 0$

$f(x) = \sin^{-1}x$

By Taylor's theorem

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!}f''(x) + \dots \quad \text{--- (1)}$$

$$f'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$f''(x) = -\frac{1 \cdot (-2x)}{2(1-x^2)^{3/2}} = \frac{x}{(1-x^2)^{3/2}}$$

Put these values in (1)

$$\sin^{-1}(x+h) = \sin^{-1}x + \frac{h}{\sqrt{1-x^2}} + \frac{h^2}{2(1-x^2)^{3/2}} + \dots$$

Q.7 If  $f(x) = x^3 + 8x^2 + 15x - 24$ , calculate the value of  $f\left(\frac{11}{10}\right)$  by the application of Taylor's series.

$$f(x) = x^3 + 8x^2 + 15x - 24$$

By Taylor's Series

$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x)$$

Put  $x=1$  and  $h=\frac{1}{10}$

$$f\left(1+\frac{1}{10}\right) = f(1) + \left(\frac{1}{10}\right) f'(1) + \left(\frac{1}{10}\right)^2 \frac{1}{2!} f''(1) + \left(\frac{1}{10}\right)^3 \frac{1}{3!} f'''(1)$$

$$f(x) = x^3 + 8x^2 + 15x - 24 \Rightarrow f(1) = 0 \quad \text{--- ①}$$

$$f'(x) = 3x^2 + 16x + 15 \Rightarrow f'(1) = 34$$

$$f''(x) = 6x + 16 \Rightarrow f''(1) = 22$$

$$f'''(x) = 6 \Rightarrow f'''(1) = 6$$

Put these values in ①

$$f\left(\frac{11}{10}\right) = 0 + \frac{34}{10} + \frac{1}{100} \times \frac{1}{2} \times 22 +$$

$$\frac{1}{1000} \times \frac{1}{6} \times 6 + \dots$$

$$f\left(\frac{11}{10}\right) = 3.511$$



Q.8 Calculate the approximate value of  $\sqrt{26}$  to three decimal places by Taylor's Expansion

$$\text{Let } f(x) = \sqrt{x}$$

$$\Rightarrow f(x+h) = \sqrt{x+h}$$

By Taylor's theorem

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \dots$$

$$\text{Put } x=25 \text{ and } h=1$$

$$f(25+1) = f(25) + f'(25) + \frac{1^2}{2!} f''(25) + \dots \quad \text{①}$$

$$f(x) = \sqrt{x} \quad \Rightarrow f(25) = \sqrt{25} = 5$$

$$f'(x) = \frac{1}{2\sqrt{x}} \quad \Rightarrow f'(25) = \frac{1}{10}$$

$$f''(x) = \frac{1}{2} \left(-\frac{1}{2}\right) \frac{1}{x^{3/2}} \Rightarrow f''(25) = -\frac{1}{500}$$

Put these values in ①

$$f(26) = 5 + \frac{1}{10} - \frac{1}{2 \times 500} + \dots$$

$$\boxed{\sqrt{26} = 5.099 \text{ approx}}$$



UNIT -3 : DPP - 2

Topic : Taylor's Theorems / Taylor's series

Q.1 Expand  $4x^2 + 7x + 5$  in powers of  $x - 3$

Q.2 Expand  $\log \sin x$  in powers of  $(x - 2)$

Q.3 Apply Taylor's Theorem to find the expansion of  $\log \sin (x + h)$

Q.4  $f(x) = x^3 - 2x + 5$ , find the value of  $f(2.001)$  with the help of Taylor's Theorem

## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 3

##### Today's target

- Taylor's Theorems and Maclaurin's Theorems  
(For the function of two Variable)
- PYQs
- DPP



By Gulshan sir



Let  $f(x, y)$  is a function of two variables

### Partial Derivative of first order

$$(i) \frac{\partial f}{\partial x} = f_x$$

$$(ii) \frac{\partial f}{\partial y} = f_y$$

### Partial Derivative of second order

$$(i) \frac{\partial^2 f}{\partial x^2} = f_{xx}$$

$$(ii) \frac{\partial^2 f}{\partial x \partial y} = f_{xy}$$

$$(iii) \frac{\partial^2 f}{\partial y^2} = f_{yy}$$

### Partial Derivative of third order

$$(i) \frac{\partial^3 f}{\partial x^3} = f_{xxx}$$

$$(ii) \frac{\partial^2 f}{\partial x^2 \partial y} = f_{xxy}$$

$$(iii) \frac{\partial^2 f}{\partial x \partial y^2} = f_{xyy}$$

$$(iv) \frac{\partial^3 f}{\partial y^3} = f_{yyy}$$

## Taylor's Theorem (Taylor's series) : For the function of two variables.

A differentiable function  $f(x, y)$  can be expanded about  $(a, b)$  by the Taylor's series

OR

A differentiable function  $f(x, y)$  can be expanded in the power of  $(x - a)$  and  $(y - b)$  by the Taylor's series

$$\begin{aligned} f(x, y) = & f(a, b) + [(x - a)f_x(a, b) + (y - b)f_y(a, b)] \\ & + \frac{1}{2!} [(x - a)^2 f_{xx}(a, b) + 2(x - a)(y - b)f_{xy}(a, b) + (y - b)^2 f_{yy}(a, b)] \\ & + \frac{1}{3!} [(x - a)^3 f_{xxx}(a, b) + 3(x - a)^2 (y - b)f_{xxy}(a, b) + 3(x - a)(y - b)^2 f_{xyy}(a, b) + (y - b)^3 f_{yyy}(a, b)] \\ & + \dots \end{aligned}$$



**Maclaurin's Theorem (Maclaurin's series) :** For the function of two variables.

A differentiable function  $f(x, y)$  can be expanded about  $(0, 0)$  by Maclaurin's series

OR

A differentiable function  $f(x, y)$  can be expanded in the power of  $x$  and  $y$  by the Maclaurin's series

$$f(x, y) = f(0, 0) + [xf_x(0, 0) + yf_y(0, 0)] + \frac{1}{2!} [x^2 f_{xx}(0, 0) + 2xy f_{xy}(0, 0) + y^2 f_{yy}(0, 0)]$$

$$+ \frac{1}{3!} [x^3 f_{xxx}(0, 0) + 3x^2 y f_{xxy}(0, 0) + 3xy^2 f_{xyy}(0, 0) + y^3 f_{yyy}(0, 0)] + \dots$$

Q. 1 Expand  $f(x, y) = e^x \cos y$  about the point  $(1, \frac{\pi}{4})$  by the Taylor's series

AKTU 2024

$$f(x, y) = e^x \cos y \Rightarrow f(1, \frac{\pi}{4}) = \frac{e}{\sqrt{2}}$$

$$f_x(x, y) = e^x \cos y \Rightarrow f_x(1, \frac{\pi}{4}) = \frac{e}{\sqrt{2}}$$

$$f_y(x, y) = -e^x \sin y \Rightarrow f_y(1, \frac{\pi}{4}) = -\frac{e}{\sqrt{2}}$$

$$f_{xx}(x, y) = e^x \cos y \Rightarrow f_{xx}(1, \frac{\pi}{4}) = \frac{e}{\sqrt{2}}$$

$$f_{xy}(x, y) = -e^x \sin y \Rightarrow f_{xy}(1, \frac{\pi}{4}) = -\frac{e}{\sqrt{2}}$$

$$f_{yy}(x, y) = -e^x \cos y \Rightarrow f_{yy}(1, \frac{\pi}{4}) = -\frac{e}{\sqrt{2}}$$

$$f_{xxx}(x, y) = e^x \cos y \Rightarrow f_{xxx}(1, \frac{\pi}{4}) = \frac{e}{\sqrt{2}}$$

$$f_{xxy}(x, y) = -e^x \sin y \Rightarrow f_{xxy}(1, \frac{\pi}{4}) = -\frac{e}{\sqrt{2}}$$

$$f_{xyy}(x, y) = -e^x \cos y \Rightarrow f_{xyy}(1, \frac{\pi}{4}) = -\frac{e}{\sqrt{2}}$$

$$f_{yyy}(x, y) = e^x \sin y \Rightarrow f_{yyy}(1, \frac{\pi}{4}) = \frac{e}{\sqrt{2}}$$



### Taylor's Theorem about (a, b)

$$f(x, y) = f(a, b) + [(x-a)f_x(a, b) + (y-b)f_y(a, b)] + \frac{1}{2!}[(x-a)^2 f_{xx}(a, b) + 2(x-a)(y-b)f_{xy}(a, b) + (y-b)^2 f_{yy}(a, b)] \\ + \frac{1}{3!}[(x-a)^3 f_{xxx}(a, b) + 3(x-a)^2(y-b)f_{xxy}(a, b) + 3(x-a)(y-b)^2 f_{xyy}(a, b) + (y-b)^3 f_{yyy}(a, b) + \dots]$$

### Taylor's Theorem about (1, $\frac{\pi}{4}$ )

$$f(x, y) = f(1, \frac{\pi}{4}) + [(x-1)f_x(1, \frac{\pi}{4}) + (y-\frac{\pi}{4})f_y(1, \frac{\pi}{4})] + \frac{1}{2!}[(x-1)^2 f_{xx}(1, \frac{\pi}{4}) + 2(x-1)(y-\frac{\pi}{4})f_{xy}(1, \frac{\pi}{4}) + (y-\frac{\pi}{4})^2 f_{yy}(1, \frac{\pi}{4})] \\ + \frac{1}{3!}[(x-1)^3 f_{xxx}(1, \frac{\pi}{4}) + 3(x-1)^2(y-\frac{\pi}{4})f_{xxy}(1, \frac{\pi}{4}) + 3(x-1)(y-\frac{\pi}{4})^2 f_{xyy}(1, \frac{\pi}{4}) + (y-\frac{\pi}{4})^3 f_{yyy}(1, \frac{\pi}{4}) + \dots]$$

$$f(x, y) = \frac{e}{\sqrt{2}} + (x-1)\frac{e}{\sqrt{2}} - (y-\frac{\pi}{4})\frac{e}{\sqrt{2}} + \frac{1}{2}(x-1)^2\frac{e}{\sqrt{2}} - (x-1)(y-\frac{\pi}{4})\frac{e}{\sqrt{2}} - \frac{1}{2}(y-\frac{\pi}{4})^2\frac{e}{\sqrt{2}} + \frac{1}{6}(x-1)^3\frac{e}{\sqrt{2}} - \frac{1}{2}(x-1)(y-\frac{\pi}{4})^2\frac{e}{\sqrt{2}} \\ + \frac{1}{6}(y-\frac{\pi}{4})^3\frac{e}{\sqrt{2}} + \dots$$

$$e^x \cos y = \frac{e}{\sqrt{2}} \left[ 1 + (n-1) - \left(y - \frac{\pi}{4}\right) + \frac{1}{2}(n-1)^2 - (n-1)\left(y - \frac{\pi}{4}\right) - \frac{1}{2}\left(y - \frac{\pi}{4}\right)^2 + \frac{1}{6}(n-1)^3 - \frac{1}{2}(n-1)^2\left(y - \frac{\pi}{4}\right) - \frac{1}{2}(n-1)\left(y - \frac{\pi}{4}\right)^2 + \frac{1}{6}\left(y - \frac{\pi}{4}\right)^3 + \dots \right]$$



about the point  $(1, 2)$ 

AKTU 2017, 2018

$$f(x, y) = x^2 + 3y^2 - 9x - 9y + 26$$

$$\Rightarrow f(1, 2) = 1 + 12 - 9 - 18 + 26 = 12$$

$$f_x(x, y) = 2x - 9$$

$$\Rightarrow f_x(1, 2) = -7$$

$$f_y(x, y) = 6y - 9$$

$$\Rightarrow f_y(1, 2) = 3$$

$$f_{xx}(x, y) = 2$$

$$\Rightarrow f_{xx}(1, 2) = 2$$

$$f_{xy}(x, y) = 0$$

$$\Rightarrow f_{xy}(1, 2) = 0$$

$$f_{yy}(x, y) = 6$$

$$\Rightarrow f_{yy}(1, 2) = 6$$

## Taylor's Series at (1, 2)

$$f(x, y) = f(1, 2) + [(x-1)f_x(1, 2) + (y-2)f_y(1, 2)] + \frac{1}{2!} [(x-1)^2 f_{xx}(1, 2) + 2(x-1)(y-2)f_{xy}(1, 2) + (y-2)^2 f_{yy}(1, 2)] + \dots$$

$$x^2 + 3y^2 - 9x - 9y + 26 = 12 - 7(x-1) + 3(y-2) + (x-1)^2 + 3(y-2)^2 + \dots$$



Q. 3 Expand  $f(x, y) = e^x \tan^{-1} y$  in powers of  $(x - 1)$  and  $(y - 1)$  up to two terms of degree 2

$$f(x, y) = e^x \tan^{-1} y$$

$$f_x(x, y) = e^x \tan^{-1} y$$

$$f_y(x, y) = \frac{e^x}{1+y^2}$$

$$f_{xx}(x, y) = e^x \tan^{-1} y$$

$$f_{xy}(x, y) = \frac{e^x}{1+y^2}$$

$$f_{yy}(x, y) = \frac{-e^x \times 2y}{(1+y^2)^2}$$

$$\Rightarrow f(1, 1) = e \times \frac{\pi}{4}$$

$$\Rightarrow f_x(1, 1) = e \times \frac{\pi}{4}$$

$$\Rightarrow f_y(1, 1) = \frac{e}{2}$$

$$\Rightarrow f_{xx}(1, 1) = e \times \frac{\pi}{4}$$

$$\Rightarrow f_{xy}(1, 1) = \frac{e}{2}$$

$$\Rightarrow f_{yy}(1, 1) = -\frac{e}{2}$$

## Taylor's Series at (1,1)

$$f(x, y) = f(1, 1) + [(x-1)f'_x(1, 1) + (y-1)f'_y(1, 1)] + \frac{1}{2}[(x-1)^2f''_{xx}(1, 1) + 2(x-1)(y-1)f''_{xy}(1, 1) + (y-1)^2f''_{yy}(1, 1)] + \dots$$

$$e^x \tan^{-1} y = \frac{\pi e}{4} + (x-1)\frac{\pi e}{4} + (y-1)\frac{e}{2} + (x-1)^2\frac{\pi e}{8} + (x-1)(y-1)\frac{e}{2} - (y-1)^2\frac{e}{4} + \dots$$



**Q.4** Expand  $x^y$  in powers of  $(x-1)$  and  $(y-1)$  up to the third degree terms and hence

evaluate  $(1.1)^{1.02}$

$$f(x, y) = x^y$$

$$\Rightarrow f(1, 1) = 1$$

$$f_x(x, y) = yx^{y-1}$$

$$\Rightarrow f_x(1, 1) = 1$$

$$f_y(x, y) = x^y \log x$$

$$\Rightarrow f_y(1, 1) = 0$$

$$f_{xx}(x, y) = y(y-1)x^{y-2}$$

$$\Rightarrow f_{xx}(1, 1) = 0$$

$$f_{xy}(x, y) = x^{y-1} + yx^{y-1} \log x \Rightarrow f_{xy}(1, 1) = 1$$

$$f_{yy}(x, y) = x^y (\log x)^2$$

$$\Rightarrow f_{yy}(1, 1) = 0$$

$$f_{xxx}(x, y) = y(y-1)(y-2)x^{y-3}$$

**AKTU-2022**

$$\Rightarrow f_{xxx}(1, 1) = 0$$

$$f_{xxy}(x, y) = (y-1)x^{y-2} + yx^{y-2} + y(y-1)x^{y-2} \log x$$

$$\Rightarrow f_{xxy}(1, 1) = 1$$

$$f_{xyy}(x, y) = 2x^{y-1} \log x + yx^{y-1} (\log x)^2$$

$$\Rightarrow f_{xyy}(1, 1) = 0$$

$$f_{yyy}(x, y) = x^y (\log x)^3$$

$$\Rightarrow f_{yyyy}(1,1) = 0$$

Taylor's Series about (1,1)

$$f(x, y) = f(1, 1) + \left[ (x-1)f_x(1, 1) + (y-1)f_y(1, 1) \right] + \frac{1}{2!} \left[ (x-1)^2 f_{xx}(1, 1) + 2(x-1)(y-1)f_{xy}(1, 1) + (y-1)^2 f_{yy}(1, 1) \right] + \frac{1}{3!} \left[ (x-1)^3 f_{xxx}(1, 1) + 3(x-1)^2(y-1)f_{xxy}(1, 1) + 3(x-1)(y-1)^2 f_{xyy}(1, 1) + (y-1)^3 f_{yyy}(1, 1) \right] + \dots$$

$$x^y = 1 + (x-1) + 0 + 0 + (x-1)(y-1) + 0 + 0 + \frac{1}{2}(x-1)^2(y-1) + 0 + 0 + \dots$$

$$x^y = 1 + (x-1) + (x-1)(y-1) + \frac{1}{2}(x-1)^2(y-1) + \dots$$



**Q.5** Find the first six terms of the expansions of the function  $e^x \log(1+y)$  in a Taylor's series in the neighbourhood of the point  $(0, 0)$ .

**AKTU-2015**

**OR**

Expand  $e^x \log(1+y)$  in power of  $x$  and  $y$  as far as the terms of third degree

$$\text{Let } f(x, y) = e^x \log(1+y) \Rightarrow f(0, 0) = 0$$

$$f_x(x, y) = e^x \log(1+y) \Rightarrow f_x(0, 0) = 0$$

$$f_y(x, y) = \frac{e^x}{1+y} \Rightarrow f_y(0, 0) = 1$$

$$f_{xx}(x, y) = e^x \log(1+y) \Rightarrow f_{xx}(0, 0) = 0$$

$$f_{xy}(x, y) = \frac{e^x}{1+y} \Rightarrow f_{xy}(0, 0) = 1$$

$$f_{yy}(x, y) = \frac{-e^x}{(1+y)^2} \Rightarrow f_{yy}(0, 0) = -1$$

$$f_{xxx}(x, y) = e^x \log(1+y) \Rightarrow f_{xxx}(0, 0) = 0$$

$$f_{xxy}(x, y) = \frac{e^x}{1+y} \Rightarrow f_{xxy}(0, 0) = 1$$

$$f_{xyy}(x, y) = \frac{-e^x}{(1+y)^2} \Rightarrow f_{xyy}(0, 0) = -1$$

$$f_{yyy}(x, y) = \frac{+2e^x}{(1+y)^3} \Rightarrow f_{yyy}(0, 0) = 2$$

## Taylor's Series about (0,0)

$$f(x,y) = f(0,0) + [x f_x(0,0) + y f_y(0,0)] + \frac{1}{2!} [x^2 f_{xx}(0,0) + 2xy f_{xy}(0,0) + y^2 f_{yy}(0,0)] + \frac{1}{3!} [x^3 f_{xxx}(0,0) + 3x^2 y f_{xxy}(0,0) + 3xy^2 f_{xyy}(0,0) + y^3 f_{yyy}(0,0)] + \dots$$

$$e^x \log(1+y) = 0 + 0 + y + 0 + xy - \frac{1}{2}y^2 + 0 + \frac{1}{2}x^2y - \frac{1}{2}xy^2 + \frac{1}{3}y^3 + \dots$$

$$e^x \log(1+y) = y + xy - \frac{1}{2}y^2 + \frac{1}{2}x^2y - \frac{1}{2}xy^2 + \frac{1}{3}y^3 + \dots$$



UNIT -3 : DPP - 3

Topic : Taylor's Theorems and Maclaurin's Theorems

Q.1 Expand  $e^x \sin y$  in powers of  $x$  and  $y$  as far as the terms of third degree

Q.2 Expand  $(x^2 y + \sin y + e^x)$  in powers of  $(x - 1)$  and  $(y - \pi)$

Q.3 Expand  $\sin xy$  in powers of  $(x - 1)$  and  $(y - \frac{\pi}{2})$  up to second degree terms

Q.4 Expand  $f(x, y) = y^x$  about  $(1, 1)$  upto second degree terms and hence evaluate  $(1.02)^{1.03}$

AKTU-2023

Q.5 Expand  $\tan^{-1} \frac{y}{x}$  in the neighbourhood of  $(1, 1)$  upto and inclusive of second degree terms.

Hence compute  $f(1.1, 0.9)$  approximately

## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 4

#### Today's target

- Maxima and Minima : For the function of two Variable
- PYQs
- DPP



By Gulshan sir



## Maxima and Minima

Function of one Variable



Function of two Variable



Function of three or more

Variables

By Lagrange's method of multipliers

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Gateway Classes

## Maxima and Minima of function of two variables

Let  $f(x, y)$  be a given function

### Steps to find Extreme values (Maximum or Minimum values)

✓ **STEP – 1 :** Partially differentiate  $f(x, y)$ , to find

$$p = \frac{\partial f}{\partial x}$$

$$q = \frac{\partial f}{\partial y}$$

$$r = \frac{\partial^2 f}{\partial x^2}$$

$$s = \frac{\partial^2 f}{\partial x \partial y}$$

$$t = \frac{\partial^2 f}{\partial y^2}$$

✓ **STEP – 2 :-** Put  $\frac{\partial f}{\partial x} = 0$  ————— (1)

$$\frac{\partial f}{\partial y} = 0 \text{ ————— (2)}$$

Solve above equations to find stationary points



**STEP – 3 :** At stationary point,

**CASE – 1 :** If  $rt - s^2 > 0$  and  $r < 0$ , then

$f(x, y)$  has maximum value at stationary point (point of maxima)

**CASE – 2 :** If  $rt - s^2 > 0$  and  $r > 0$ , then

$f(x, y)$  has minimum value at stationary point (point of minima)

**CASE – 3 :** If  $rt - s^2 < 0$ , then

$f(x, y)$  has neither maximum nor minimum at stationary point (Saddle point)

**CASE – 4 :** If  $rt - s^2 = 0$ , then case is doubtful

**STEP – 5 :** Find the maximum and minimum values of at stationary points

**Q.1** Find the extreme values of function :  $x^2 + y^2 + 6x + 12$

$$\text{Let } f(x, y) = x^2 + y^2 + 6x + 12$$

$$\frac{\partial f}{\partial x} = 2x + 6$$

$$\frac{\partial f}{\partial y} = 2y$$

$$r = \frac{\partial^2 f}{\partial x^2} = 2$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$t = \frac{\partial^2 f}{\partial y^2} = 2$$

Put

$$\frac{\partial f}{\partial x} = 0$$

$$2x + 6 = 0$$

$$\boxed{x = -3}$$

$$\frac{\partial f}{\partial y} = 0$$

$$2y = 0$$

$$\boxed{y = 0}$$

At Stationary point  $(-3, 0)$

$$rt - s^2 = 2 \times 2 - 0^2 = 4$$

$$\Rightarrow rt - s^2 > 0 \text{ and } r > 0$$

$\Rightarrow f(x, y)$  has minimum value at  $(-3, 0)$

Minimum value

$$f(x, y) = x^2 + y^2 + 6x + 12$$

$$\begin{aligned} f(-3, 0) &= (-3)^2 + 0^2 + 6(-3) + 12 \\ &= 9 - 18 + 12 \end{aligned}$$

$$\boxed{f(-3, 0) = 3}$$

Point of  
minima



**Q.2** Find the extreme values of function :  $x^3 + y^3 - 63(x + y) + 12xy$

Let  $f(x, y) = x^3 + y^3 - 63(x + y) + 12xy$

$$\frac{\partial f}{\partial x} = 3x^2 - 63 + 12y$$

$$\frac{\partial f}{\partial y} = 3y^2 - 63 + 12x$$

$$r = \frac{\partial^2 f}{\partial x^2} = 6x$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = 12$$

$$t = \frac{\partial^2 f}{\partial y^2} = 6y$$

Put

$$\frac{\partial f}{\partial x} = 0$$

$$3x^2 - 63 + 12y = 0$$

$$3x^2 + 12y = 63$$

$$3(x^2 + 4y) = 63$$

$$x^2 + 4y = 21 \quad \text{--- (1)}$$

$$\frac{\partial f}{\partial y} = 0$$

$$3y^2 - 63 + 12x = 0$$

$$3y^2 + 12x = 63$$

$$y^2 + 4x = 21 \quad \text{--- (2)}$$

From (1) and (2)

$$x^2 + 4y = y^2 + 4x$$

$$x^2 - y^2 + 4y - 4x = 0$$

$$(x - y)(x + y) - 4(x - y) = 0$$

$$(x - y)(x + y - 4) = 0$$

$$\boxed{y = x}$$

$$x + y = 4$$

$$\boxed{y = 4 - x}$$

CASE-1: When  $y = x$

From ①

$$x^2 + 4y = 21$$

$$x^2 + 4x = 21$$

$$x^2 + 4x - 21 = 0$$

$$x = 3, -7$$

Stationary points

$(3, 3)$  and  $(-7, -7)$

CASE-2: When  $y = 4 - x$

From ①

$$x^2 + 4y = 21$$

$$x^2 + 4(4 - x) = 21$$

$$x^2 + 16 - 4x = 21$$

$$x^2 - 4x - 5 = 0$$

$$x = -1, 5$$

When  $x = -1$ ,  $y = 5$

When  $x = 5$ ,  $y = -1$

Two more Stationary points

$(-1, 5)$  and  $(5, -1)$

At Stationary point  $(3, 3)$

$$x_t - s^2 = 6x \times 6y - 12^2$$

$$x_t - s^2 = 6 \times 3 \times 6 \times 3 - 12 \times 12$$

$$x_t - s^2 = 180 > 0$$

$$x = 6x = 6 \times 3 = 18 > 0$$

$$\Rightarrow x_t - s^2 > 0 \text{ and } x > 0$$

$\Rightarrow f(x, y)$  has minimum value at

$(3, 3) \rightarrow$  point of minima



$$f(x, y) = x^3 + y^3 - 63(x+y) + 12xy$$

$$f(3, 3) = (3)^3 + (3)^3 - 63(3+3) + 12 \times 3 \times 3$$

$$= 27 + 27 - 63 \times 6 + 108$$

$$f(3, 3) = -216$$

At Stationary point  $(-7, -7)$

$$r - s^2 = 6x \times 6y - 12^2$$

$$= 6(-7) \times 6(-7) - 12^2$$

$$= 42^2 - 12^2$$

$$= 1620 > 0$$

$$r = 6x = 6(-7)$$

$$r = -42 < 0$$

$\Rightarrow f(x, y)$  has maximum value at  $(-7, -7)$

Point of maxima

$$f(x, y) = x^3 + y^3 - 63(x+y) + 12xy$$

$$f(-7, -7) = (-7)^3 + (-7)^3 - 63(-7-7) + 12(-7)(-7)$$

$$= -343 - 343 + 882 + 588$$

$$f(-7, -7) = 784$$

At stationary point  $(-1, 5)$

$$rt - s^2 = 6x \times 6y - 12^2$$

$$= 6(-1) \times 6 \times 5 - 144$$

$$= -180 - 144$$

$$= -324 < 0$$

$$\Rightarrow rt - s^2 < 0$$

$\Rightarrow f(x, y)$  has neither maximum  
nor minimum at  $(-1, 5)$

Saddle point

At stationary point  $(5, -1)$

$$rt - s^2 = 6x \times 6y - 12^2$$

$$= 6 \times 5 \times 6 \times (-1) - 12^2$$

$$= -180 - 144$$

$$= -324 < 0$$

$$\Rightarrow rt - s^2 < 0$$

$\Rightarrow f(x, y)$  has neither maximum nor minimum

at  $(5, -1)$

Saddle point



**Q.3** Examine for minimum and maximum values :  $\sin x + \sin y + \sin(x + y)$ .

Let  $f(x, y) = \sin x + \sin y + \sin(x + y)$

$$\frac{\partial f}{\partial x} = \cos x + \cos(x + y)$$

$$\frac{\partial f}{\partial y} = \cos y + \cos(x + y)$$

$$r = \frac{\partial^2 f}{\partial x^2} = -\sin x - \sin(x + y)$$

$$s = \frac{\partial^2 f}{\partial x \partial y} = -\sin(x + y)$$

$$t = \frac{\partial^2 f}{\partial y^2} = -\sin y - \sin(x + y)$$

Put

$$\frac{\partial f}{\partial x} = 0$$

$$\cos x + \cos(x + y) = 0 \quad \text{--- (1)}$$

$$\frac{\partial f}{\partial y} = 0$$

$$\cos y + \cos(x + y) = 0 \quad \text{--- (2)}$$

Subtract eq<sup>n</sup> (2) from (1)

$$\cos x + \cancel{\cos(x + y)} - (\cos y + \cancel{\cos(x + y)}) = 0$$

$$\cos x = \cos y$$

$$\Rightarrow x = y$$

Put  $y = x$  in (1)

$$\cos x + \cos(x+x) = 0$$

$$\cos x + \cos 2x = 0$$

$$\cos 2x = -\cos x$$

$$\cos 2x = \cos(\pi - x)$$

$$2x = \pi - x$$

$$3x = \pi$$

$$x = \frac{\pi}{3}$$

$$y = \frac{\pi}{3}$$

At Stationary point  $(\frac{\pi}{3}, \frac{\pi}{3})$

$$x = -\sin x - \sin(x+y)$$

$$x = -\sin \frac{\pi}{3} - \sin \frac{2\pi}{3}$$

$$x = -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}$$

$$x = -\sqrt{3}$$

$$s = -\sin(x+y)$$

$$s = -\sin \frac{2\pi}{3}$$

$$s = -\frac{\sqrt{3}}{2}$$

$$t = -\sin y - \sin(x+y)$$

$$= -\sin \frac{\pi}{3} - \sin \frac{2\pi}{3}$$

$$= -\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2}$$

$$t = -\sqrt{3}$$

Now

$$xt - s^2 = (-\sqrt{3})(-\sqrt{3}) - \left(-\frac{\sqrt{3}}{2}\right)^2$$

$$= 3 - \frac{3}{4}$$

$$= \frac{12-9}{4} = \frac{3}{4}$$



$$rt - s^2 = \frac{3}{4} > 0$$

$$r = -\sqrt{3} < 0$$

$$\Rightarrow rt - s^2 > 0 \text{ and } r < 0$$

$$\Rightarrow f(x, y) \text{ has maximum value at } \left(\frac{\pi}{3}, \frac{\pi}{3}\right)$$

Maximum value

$$f(x, y) = \sin x + \sin y + \sin(x+y)$$

$$f\left(\frac{\pi}{3}, \frac{\pi}{3}\right) = \sin \frac{\pi}{3} + \sin \frac{\pi}{3} + \sin 2\frac{\pi}{3}$$

$$= \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}$$

Point of maxima

$$f\left(\frac{\pi}{3}, \frac{\pi}{3}\right) = \frac{3\sqrt{3}}{2}$$

## UNIT -3 : DPP - 4

Topic : Maxima and Minima : For the function of two Variable

- Q.1 Find the extreme values of function  $x^3 + y^3 - 3axy$
- Q.2 In a plane triangle ABC, find the maximum value of  $\cos A \cos B \cos C$ .
- Q.3 A rectangular box, open at the top, is to have a given capacity. Find the dimensions of the box requiring least material for its construction.



## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 5

#### Today's target

- Lagrange's method of multipliers (*Part – 1*)  
(Maxima and Minima for the function of three variables)
- PYQs
- DPP



By Gulshan sir

## Maxima and Minima

Function of one Variable

X

Function of two Variable

Function of three or more

Variables

By Lagrange's method of multipliers

7

Gateway Classes



This method is used to find maximum and minimum value of a function of three or more variables in which all the variables are connected by a given relation.

### Steps to solve questions

**STEP – 1 :** Identify the Function (Maximum or Minimum) and Given Relation

(i) Let  $f(x, y, z) =$  Function

(ii) Let  $\phi(x, y, z) =$  Given Relation

**STEP – 2 :** Consider the Lagrange's function

$$F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$$

$$dF = \left( \frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x} \right) dx + \left( \frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y} \right) dy + \left( \frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z} \right) dz$$

Known as Lagrange's identity

For Maxima and Minima

$$dF = 0$$

$$\frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0 \quad \text{--- (3)}$$

$$\frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0 \quad \text{--- (4)}$$

$$\frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0 \quad \text{--- (5)}$$

**STEP - 3 :** Solve equations 3, 4 and 5 to find  $x, y$  and  $z$

(You can take the help of given Relation)

**STEP - 4 :** Put  $x, y$  and  $z$  in the given function to get maximum and minimum value



**Q.1** A rectangular box, which is open at the top, has a capacity of **32 c.c.** Determine, using Lagrange's method of multipliers, the dimensions of the box such the least material is required for the construction of the box.

**AKTU 2022, 2023**

Let  $x, y, z$  be length, breadth and height of the box

$$S = xy + 2yz + 2zx \text{ ————— ①}$$

$$f(x, y, z) = xy + 2yz + 2zx$$

$$V = 32$$

$$xyz = 32 \text{ ————— ②}$$

$$\phi(x, y, z) = xyz - 32$$

Consider Lagrange's Function.

$$F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$$

$$F(x, y, z) = xy + 2yz + 2zx + \lambda(xyz - 32)$$

$$dF = (y + 2z + \lambda yz)dx + (x + 2z + \lambda xz)dy + (2y + 2x + \lambda xy)dz$$

For Maxima and Minima

$$dF = 0$$

$$y + 2z + \lambda yz = 0 \Rightarrow -\lambda = \frac{y+2z}{yz} \quad \text{--- (3)}$$

$$x + 2z + \lambda xz = 0 \Rightarrow -\lambda = \frac{x+2z}{xz} \quad \text{--- (4)}$$

$$2y + 2x + \lambda xy = 0 \Rightarrow -\lambda = \frac{2y+2x}{xy} \quad \text{--- (5)}$$

Combined eq<sup>n</sup> (3), (4) and (5)

$$\frac{y+2z}{yz} = \frac{x+2z}{xz} = \frac{2y+2x}{xy}$$

(i)                      (ii)                      (iii)

$$\frac{y+2z}{y\cancel{z}} = \frac{x+2z}{x\cancel{z}}$$

$$\cancel{xy} + 2xz = \cancel{xy} + 2yz$$

$$\cancel{2xz} = \cancel{2yz}$$

$$\boxed{x=y} \quad \text{--- (6)}$$

$$\frac{x+2z}{x\cancel{z}} = \frac{2y+2x}{x\cancel{y}}$$

$$xy + 2y\cancel{z} = 2y\cancel{z} + 2zx$$

$$\cancel{xy} = 2zx\cancel{x}$$

$$\boxed{y=2z} \quad \text{--- (7)}$$



combined ⑥ and ⑦

$$x = y = 2z$$

$$\frac{x}{2} = \frac{y}{2} = \frac{z}{1} = k \text{ (say)}$$

$$x = 2k$$

$$y = 2k$$

$$z = k$$

Put  $x, y, z$  in ②

$$2k \times 2k \times k = 32$$

$$k^3 = \frac{32}{4}$$

$$k^3 = 8$$

$$k = 2$$

Hence

$$\begin{aligned} x &= 4 \\ y &= 4 \\ z &= 2 \end{aligned}$$

**Q.2** Show that the rectangular solid of maximum volume that can be inscribed in given sphere is a cube. Also find maximum volume

Let  $2x, 2y, 2z$  be the length, breadth and height of rectangular solid

Let  $R$  be the Radius of sphere

$$V = 2x \times 2y \times 2z$$

$$V = 8xyz \quad \text{————— ①}$$

$$f(x, y, z) = 8xyz$$

Equation of sphere

$$x^2 + y^2 + z^2 = R^2 \quad \text{————— ②}$$

$$\phi(x, y, z) = x^2 + y^2 + z^2 - R^2$$

Consider Lagrange's Function

$$F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$$

$$F(x, y, z) = 8xyz + \lambda(x^2 + y^2 + z^2 - R^2)$$

$$dF = (8yz + 2\lambda x)dx + (8xz + 2\lambda y)dy + (8xy + 2\lambda z)dz$$

For Maxima and Minima

$$dF = 0$$

$$8yz + 2\lambda x = 0 \quad \text{————— ③}$$

$$8xz + 2\lambda y = 0 \quad \text{————— ④}$$

$$8xy + 2\lambda z = 0 \quad \text{————— ⑤}$$



Multiplying (3) by  $x$ , (4) by  $y$  and (5) by  $z$

$$8xyz + 2\lambda x^2 = 0 \Rightarrow 8xyz = -2\lambda x^2 \text{ ——— (6)}$$

$$8xyz + 2\lambda y^2 = 0 \Rightarrow 8xyz = -2\lambda y^2 \text{ ——— (7)}$$

$$8xyz + 2\lambda z^2 = 0 \Rightarrow 8xyz = -2\lambda z^2 \text{ ——— (8)}$$

Combined (6), (7) and (8)

$$-2\cancel{\lambda}x^2 = -2\cancel{\lambda}y^2 = -2\cancel{\lambda}z^2$$

$$x^2 = y^2 = z^2$$

$$x = y = z \text{ ——— (9)}$$

$$2x = 2y = 2z$$

$$l = b = h$$

Hence

Rectangular solid is a cube

using (9) in (2)

$$x^2 + x^2 + x^2 = R^2$$

$$3x^2 = R^2$$

$$x = \frac{R}{\sqrt{3}}, \quad y = \frac{R}{\sqrt{3}} \quad \text{and} \quad z = \frac{R}{\sqrt{3}}$$

Put  $x, y, z$  in (1)

$$V = 8 \times \frac{R}{\sqrt{3}} \times \frac{R}{\sqrt{3}} \times \frac{R}{\sqrt{3}} = \frac{8R^3}{3\sqrt{3}}$$

Q.3 The pressure  $P$  at any point  $(x, y, z)$  in space is  $P = 400xyz^2$ . Find the highest pressure at the surface of a unit sphere  $x^2 + y^2 + z^2 = 1$  using Lagrange's Method

AKTU 2024

$$P = 400xyz^2 \text{ ————— ①}$$

$$f(x, y, z) = 400xyz^2$$

$$x^2 + y^2 + z^2 = 1 \text{ ————— ②}$$

$$\phi(x, y, z) = x^2 + y^2 + z^2 - 1$$

Consider Lagrange's Function

$$F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$$

$$F(x, y, z) = 400xyz^2 + \lambda(x^2 + y^2 + z^2 - 1)$$

$$dF = (400yz^2 + 2\lambda x)dx + (400xz^2 + 2\lambda y)dy + (800xyz + 2\lambda z)dz$$

For Maxima and Minima

$$dF = 0$$

$$400yz^2 + 2\lambda x = 0 \text{ ————— (3)}$$

$$400xz^2 + 2\lambda y = 0 \text{ ————— (4)}$$

$$800xyz + 2\lambda z = 0 \text{ ————— (5)}$$

Multiplying (3) by  $x$ , (4) by  $y$  and (5) by  $z$



$$400\pi y^2 + 2\lambda n^2 = 0 \text{ — (6)}$$

$$400\pi yz^2 + 2\lambda y^2 = 0 \text{ — (7)}$$

$$800\pi yz^2 + 2\lambda z^2 = 0 \text{ — (8)}$$

Adding (6), (7) and (8)

$$1600\pi yz^2 + 2\lambda(n^2 + y^2 + z^2) = 0$$

$$1600\pi yz^2 + 2\lambda = 0 \text{ (using (2))}$$

$$\lambda = -800\pi yz^2$$

Put  $\lambda$  in (3)

$$400\pi yz^2 - 2 \times 800\pi yz^2 \times n = 0$$

$$400\pi yz^2 = 1600\pi n^2 yz^2$$

$$n^2 = \frac{400}{1600} 4$$

$$n = \pm \frac{1}{2}$$

Put  $\lambda$  in (4)

$$400\pi n^2 z^2 - 2 \times 800\pi yz^2 \times y = 0$$

$$400\pi n^2 z^2 = 1600\pi y^2 n^2 z^2$$

$$y^2 = \frac{400}{1600} 4$$

$$y = \pm \frac{1}{2}$$

Put  $\lambda$  in (5)

$$800\pi yz - 2 \times 800\pi yz^2 \times z = 0$$

$$800\pi yz = 1600\pi yz^3$$

$$z^2 = \frac{800}{1600} 2$$

$$z = \pm \frac{1}{\sqrt{2}}$$

Highest Pressure

$$P = 400\pi yz = 400 \times \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

$$P = 50 \text{ unit}$$

## UNIT -3 : DPP - 5

## Topic : Lagrange's method of multipliers

**Q.1** A rectangular box, which is open at the top, has a capacity of 256 cubic feet. Determine, using Lagrange's method of multipliers, the dimensions of the box such the least material is required for the construction of the box

**Q.2** Find the dimensions of rectangular box of maximum capacity whose surface area is given when

(i) Box is open at the top

(ii) Box is closed

AKTU 2016

**Q.3** Find the minimum value of  $x^2 + y^2 + z^2$ , given that  $ax + by + cz = p$ .



## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 6

#### Today's target

- Lagrange's method of multipliers (*Part – 2*)

(Maxima and Minima for the function of three variables)

- PYQs
- DPP



By Gulshan sir

This method is used to find maximum and minimum value of a function of three or more variables in which all the variables are connected by a given relation.

### Steps to solve questions

**STEP – 1 :** Identify the Function (Maximum or Minimum) and Given Relation

(i) Let  $f(x, y, z)$  = Function (Maximum or Minimum)

(ii) Let  $\phi(x, y, z)$  = Given Relation

**STEP – 2 :** Consider the Lagrange's function

$$F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$$



$$dF = \left( \frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x} \right) dx + \left( \frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y} \right) dy + \left( \frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z} \right) dz$$

Known as Lagrange's identity

For Maxima and Minima

$$dF = 0$$

$$\frac{\partial f}{\partial x} + \lambda \frac{\partial \phi}{\partial x} = 0 \quad \text{--- (3)}$$

$$\frac{\partial f}{\partial y} + \lambda \frac{\partial \phi}{\partial y} = 0 \quad \text{--- (4)}$$

$$\frac{\partial f}{\partial z} + \lambda \frac{\partial \phi}{\partial z} = 0 \quad \text{--- (5)}$$

**STEP – 3 :** Solve equations 3, 4 and 5 to find  $x, y$  and  $z$

(You can take the help of given Relation)

**STEP – 4 :** Put  $x, y$  and  $z$  in the given function to get maximum and minimum value

**Q.1** Divide 24 into three parts such that the continued product of first, square of second and cube of third may be maximum.

**AKTU 2021**

Let  $x, y, z$  be the three parts of 24

$$P = xy^2z^3 \text{ ——— (1)}$$

$$f(x, y, z) = xy^2z^3$$

$$x + y + z = 24 \text{ ——— (2)}$$

$$\phi(x, y, z) = x + y + z - 24$$

consider Lagrange's Function

$$F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$$

$$F(x, y, z) = xy^2z^3 + \lambda(x + y + z - 24)$$

$$dF = (y^2z^3 + \lambda)dx + (2xy^2z^3 + \lambda)dy + (3xy^2z^2 + \lambda)dz$$

For maxima and minima

$$dF = 0$$

$$y^2z^3 + \lambda = 0 \text{ ——— (3)}$$

$$2xy^2z^3 + \lambda = 0 \text{ ——— (4)}$$

$$3xy^2z^2 + \lambda = 0 \text{ ——— (5)}$$



From (3), (4) and (5)

$$\lambda = -y^2 z^3 \text{ --- (6)}$$

$$\lambda = -2nyz^3 \text{ --- (7)}$$

$$\lambda = -3ny^2 z^2 \text{ --- (8)}$$

Combined (6), (7) and (8)

$$+y^2 z^3 = +2nyz^3 = +3ny^2 z^2$$

From (i) and (ii)

$$y^2 \cancel{z^3} = 2ny \cancel{z^3}$$

$$y = 2n \text{ --- (9)}$$

From (ii) and (iii)

$$2n \cancel{y} \cancel{z^3} = 3n \cancel{y^2} \cancel{z^2}$$

$$2z = 3y$$

$$y = \frac{2}{3}z \text{ --- (10)}$$

combined (9) and

$$(10)$$

$$2n = y = \frac{2}{3}z$$

$$\frac{n}{1} = \frac{y}{2} = \frac{z}{3} = k$$

$$n = k$$

$$y = 2k$$

$$z = 3k$$

Put  $n, y, z$  in (2)

$$k + 2k + 3k = 24$$

$$6k = 24$$

$$k = 4$$

$$\Rightarrow x = 4$$

$$y = 8$$

$$z = 12$$

Put  $x, y, z$  in ①

$$P = xyz^2z^3$$

$$P = 4 \times 8^2 \times 12^3$$



**Q.2** Find the shortest and longest distances from the point  $(1, 2, -1)$  to the sphere  $x^2 + y^2 + z^2 = 24$

Let  $(x, y, z)$  be a point on  
sphere  $x^2 + y^2 + z^2 = 24$

$$OA = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

$$OA = \sqrt{(x-1)^2 + (y-2)^2 + (z+1)^2}$$

$$OA^2 = (x-1)^2 + (y-2)^2 + (z+1)^2$$

①

$$f(x, y, z) = (x-1)^2 + (y-2)^2 + (z+1)^2$$

$$x^2 + y^2 + z^2 = 24 \quad \text{--- ②}$$

$$\phi(x, y, z) = x^2 + y^2 + z^2 - 24$$

Consider Lagrange's function

$$F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$$

$$F(x, y, z) = (x-1)^2 + (y-2)^2 + (z+1)^2 + \lambda(x^2 + y^2 + z^2 - 24)$$

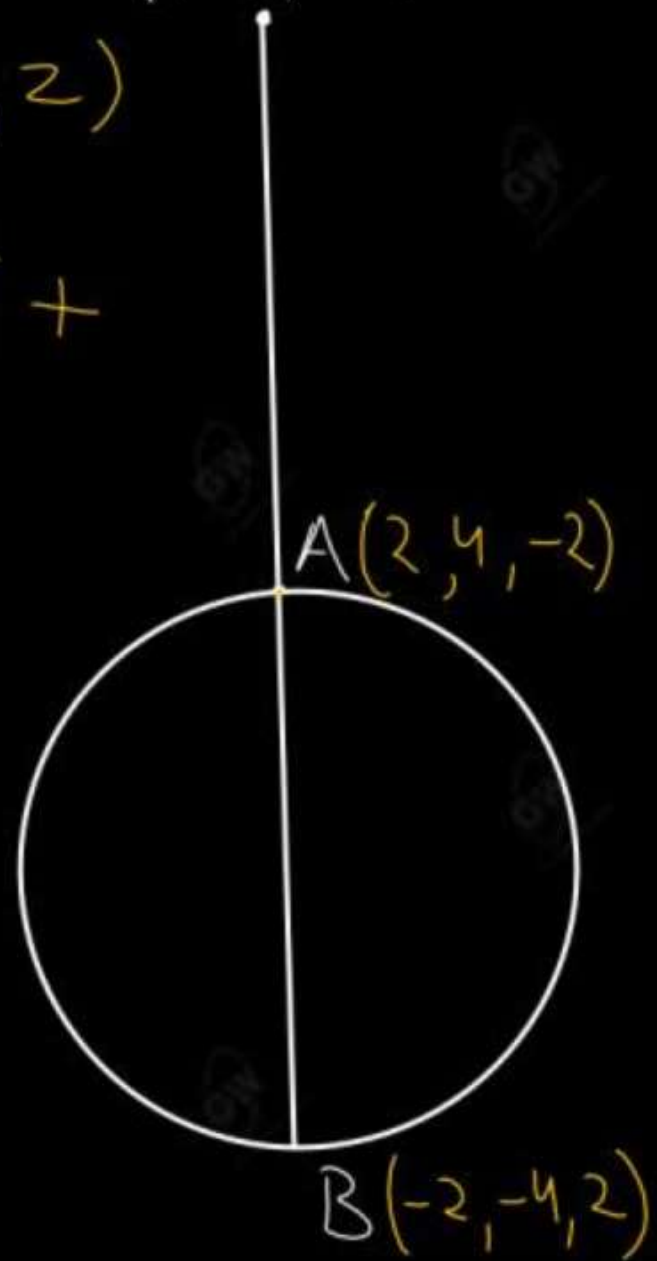
$$dF = [2(x-1) + 2\lambda x]dx +$$

$$[2(y-2) + 2\lambda y]dy +$$

$$[2(z+1) + 2\lambda z]dz$$

**AKTU 2019**

$O(1, 2, -1)$



For maxima and minima combined (6), (7) and (8)

$$dF = 0$$

$$2(n-1) + 2\lambda n = 0 \quad \text{--- (3)}$$

$$2(y-2) + 2\lambda y = 0 \quad \text{--- (4)}$$

$$2(z+1) + 2\lambda z = 0 \quad \text{--- (5)}$$

From (3), (4) and (5)

$$\lambda = -\frac{2(n-1)}{2n} \quad \text{--- (6)}$$

$$\lambda = -\frac{2(y-2)}{2y} \quad \text{--- (7)}$$

$$\lambda = -\frac{2(z+1)}{2z} \quad \text{--- (8)}$$

$$\frac{+2(n-1)}{2n} = \frac{+2(y-2)}{2y} = \frac{+2(z+1)}{2z}$$

$$\frac{n-1}{n} = \frac{y-2}{y} = \frac{z+1}{z}$$

$$1 - \frac{1}{n} = 1 - \frac{2}{y} = 1 + \frac{1}{z}$$

$$-\frac{1}{n} = -\frac{2}{y} = \frac{1}{z}$$

$$-\frac{1}{n} = -\frac{y}{2} = \frac{z}{1} = K \text{ (say)}$$

$\Rightarrow$

$$n = -K$$

$$y = -2K$$

$$z = K$$

Put  $n, y, z$  in (2)

$$(-K)^2 + (-2K)^2 + K^2 = 24$$

$$K^2 + 4K^2 + K^2 = 24$$

$$6K^2 = 24$$

$$K^2 = 4$$



$$k = \pm 2$$

$$\Rightarrow x = \mp 2$$

$$y = \mp 4$$

$$z = \pm 2$$

$$\Rightarrow A(2, 4, -2)$$

$$B(-2, -4, 2)$$

Put  $x, y, z$  in ①

$$OA = \sqrt{(2-1)^2 + (4-2)^2 + (-2+1)^2}$$

$$OA = \sqrt{1+4+1}$$

$$OA = \sqrt{6}$$

Also

$$OB = \sqrt{(-2-1)^2 + (-4-2)^2 + (2+1)^2}$$

$$OB = \sqrt{9+36+9}$$

$$OB = 3\sqrt{6}$$

Hence

$$\text{Longest distance} = 3\sqrt{6}$$

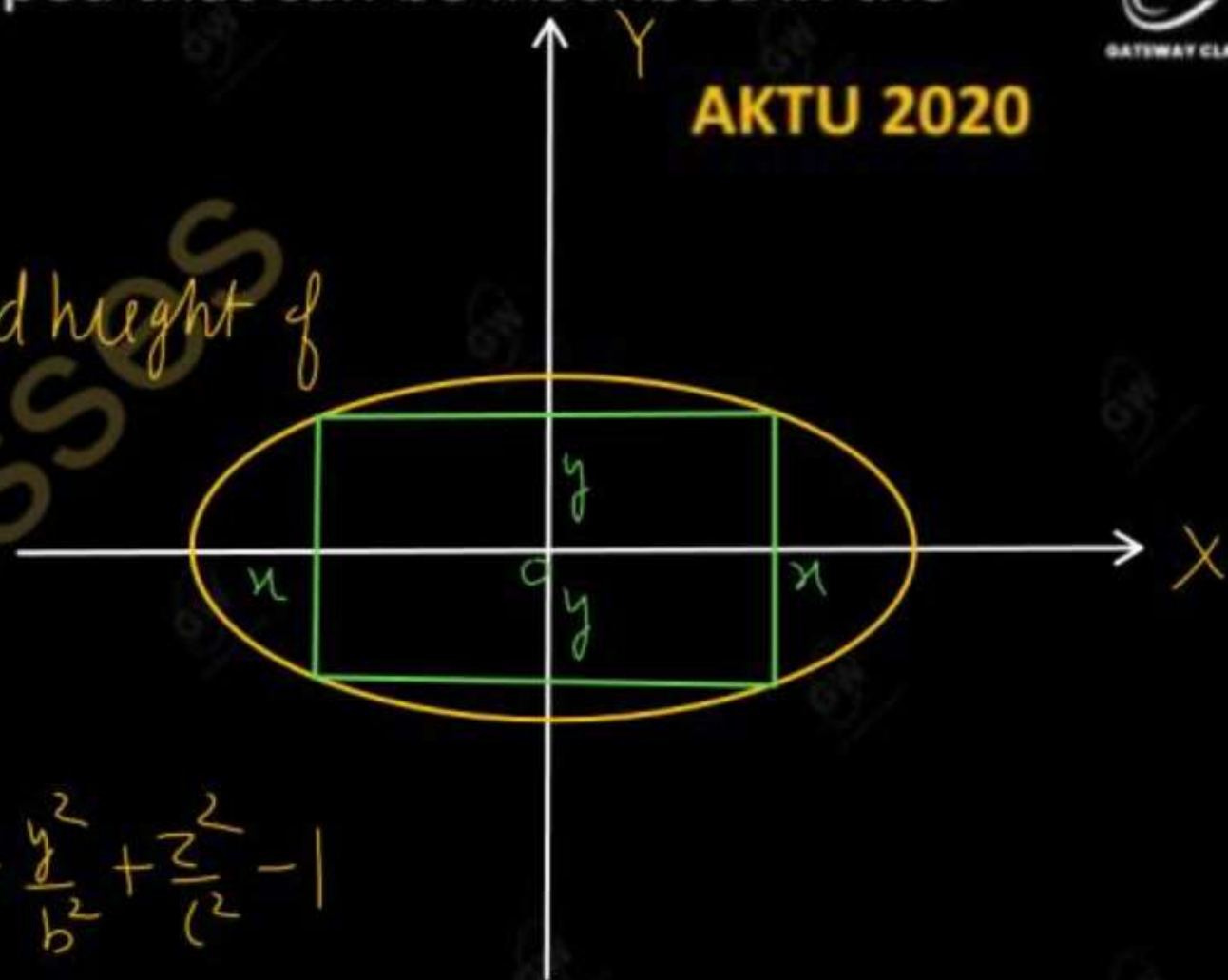
$$\text{Shortest distance} = \sqrt{6}$$

Q.3 Find the volume of the largest rectangular parallelepiped that can be inscribed in the

ellipsoid  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

AKTU 2020

Let  $2x$ ,  $2y$  and  $2z$  be the length, breadth and height of the rectangular parallelepiped



$$V = 2x \times 2y \times 2z$$

$$V = 8xyz \quad \text{--- (1)}$$

$$f(x, y, z) = 8xyz$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \text{--- (2)}$$

$$\phi(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$$

consider Lagrange function

$$F(x, y, z) = f(x, y, z) + \lambda \phi(x, y, z)$$



$$F(x, y, z) = 8xyz + \lambda \left( \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right)$$

$$dF = \left( 8yz + \frac{2\lambda x}{a^2} \right) dx + \left( 8xz + \frac{2\lambda y}{b^2} \right) dy + \left( 8xy + \frac{2\lambda z}{c^2} \right) dz$$

For maxima and minima

$$dF = 0$$

$$8yz + \frac{2\lambda x}{a^2} = 0 \quad \text{--- (3)}$$

$$8xz + \frac{2\lambda y}{b^2} = 0 \quad \text{--- (4)}$$

$$8xy + \frac{2\lambda z}{c^2} = 0 \quad \text{--- (5)}$$

Multiply (3) by  $x$ , (4) by  $y$ , (5) by  $z$  and

$$8xyz + 2\lambda \frac{x^2}{a^2} + 8xyz + 2\lambda \frac{y^2}{b^2} + 8xyz + 2\lambda \frac{z^2}{c^2} = 0$$

$$2\lambda \frac{z^2}{c^2} = 0$$

$$24xyz + 2\lambda \left( \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) = 0 \quad \text{--- (6)}$$

Using (2) in (6)

$$24xyz + 2\lambda = 0$$

$$\lambda = -12xyz$$

Put  $\lambda$  in (3)

$$8yz - \frac{2n}{a^2} \times 12nyz = 0$$

~~$$8yz = \frac{2n}{a^2} \times 12nyz$$~~

$$1 = \frac{3n^2}{a^2}$$

$$n^2 = \frac{a^2}{3}$$

$$n = \frac{a}{\sqrt{3}}$$

Similarly

$$y = \frac{b}{\sqrt{3}}$$

$$z = \frac{c}{\sqrt{3}}$$

Put  $n, y, z$  in (1)

$$V = 8nyz$$

$$V = 8 \times \frac{a}{\sqrt{3}} \times \frac{b}{\sqrt{3}} \times \frac{c}{\sqrt{3}}$$

$$V = \frac{8abc}{3\sqrt{3}}$$



UNIT -3 : DPP - 6

Topic : Lagrange's method of multipliers

**Q.1** Divide a number into three parts such that the product of first , square of second and cube of third is maximum. AKTU 2017

**Q.2** The sum of three positive numbers is constant. Prove that their product is maximum when they are equal.

**Q.3** Find the maximum and minimum distance of the point (3, 4, 12) from the sphere  $x^2 + y^2 + z^2 = 1$

## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 7

#### Today's target

- Approximation of Error
- PYQs
- DPP



By Gulshan sir



$$\textcircled{1} y = \log_{10} x$$

- Calculation of approximate error in one variable (independent or dependent variables) when error in other variables are given

✓ (i) Let  $u = f(x)$

②  $A = l \times b$

③  $V = l \times b \times h$

If  $\delta x$ , is a small increments in  $x$ , then  $\delta u$  be the corresponding increments in  $u$

$$\delta u = \frac{df}{dx} \delta x$$

✓ (ii) Let  $u = f(x, y)$

If  $\delta x$ ,  $\delta y$  are small increments in  $x$  and  $y$ , then  $\delta u$  be the corresponding increments in  $u$

$$\delta u = \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y$$

Approximately

OR

$$du = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

Total differentiation

(iii) Let  $u = f(x, y, z)$

If  $\delta x, \delta y, \delta z$  are small increments in  $x$  and  $y, z$  then  $\delta u$  be the corresponding increments in  $u$

$$\delta u = \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y + \frac{\partial f}{\partial z} \delta z$$

Approximately

OR

$$du = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

Total differentiation

**Types of error :** if  $\delta x$  is the error in  $x$ , then

(i) Absolute error =  $\delta x$

(ii) Relative error =  $\frac{\delta x}{x}$

(iii) Percentage error (%) =  $\frac{\delta x}{x} \times 100$



## TYPE – 1 : By taking logarithm

### Steps to solve questions

**STEP – 1** : Taking log both side of the function

**STEP – 2** : Differentiate above function

**STEP – 3** : Put the given value in the function

Gateway Classes

Q.1 Calculate the error in R if  $E = RI$  and possible error in E and I are 20% and 10% respectively.

AKTU 2024

$$\frac{\delta E}{E} \times 100 = 20\%$$

$$\frac{\delta I}{I} \times 100 = 10\%$$

$$\frac{\delta R}{R} \times 100 = ?$$

$$E = RI$$

Taking log both side

$$\log E = \log RI$$

$$\log E = \log R + \log I$$

$$\frac{1}{E} \delta E = \frac{1}{R} \delta R + \frac{1}{I} \delta I$$

$$\frac{\delta E}{E} \times 100 = \frac{\delta R}{R} \times 100 + \frac{\delta I}{I} \times 100$$

$$20 = \frac{\delta R}{R} \times 100 + 10\%$$

$$\frac{\delta R}{R} \times 100 = 10\%$$



**Q.2** Find the percentage error in measuring the volume of a rectangular box when the error of 1 % is made in measuring each side

**AKTU 2017, 2023**

Let  $l, b, h$  be the length, breadth and height of the box

$$\frac{\delta l}{l} \times 100 = 1\%$$

$$\frac{\delta b}{b} \times 100 = 1\%$$

$$\frac{\delta h}{h} \times 100 = 1\%$$

$$\frac{\delta V}{V} \times 100 = ?$$

$$V = l \times b \times h$$

Taking log both side

$$\log V = \log(l \times b \times h)$$

$$\log V = \log l + \log b + \log h$$

$$\frac{1}{V} \delta V = \frac{1}{l} \delta l + \frac{1}{b} \delta b + \frac{1}{h} \delta h$$

$$\frac{\delta V}{V} \times 100 = \frac{\delta l}{l} \times 100 + \frac{\delta b}{b} \times 100 + \frac{\delta h}{h} \times 100$$

$$\frac{\delta V}{V} \times 100 = 1\% + 1\% + 1\% = 3\%$$

increase in P

$$\frac{\delta u}{u} \times 100 = 3\%$$

$$\frac{\delta l}{l} \times 100 = -1\%$$

$$\frac{\delta P}{P} \times 100 = ?$$

$$P = \lambda u^3 l^3$$

Taking log both side

$$\log P = \log (\lambda u^3 l^3)$$

$$\log P = \log \lambda + \log u^3 + \log l^3$$

$$\log P = \log \lambda + 3 \log u + 3 \log l$$

$$\frac{1}{P} \delta P = 0 + \frac{3}{u} \delta u + \frac{3}{l} \delta l$$

$$\frac{\delta P}{P} \times 100 = 3 \frac{\delta u}{u} \times 100 + 3 \frac{\delta l}{l} \times 100$$

$$\frac{\delta P}{P} \times 100 = 3 \times 3\% + 3 \times (-1\%)$$

$$\frac{\delta P}{P} \times 100 = 6\%$$



**Q.4** The period of a simple pendulum is  $T = 2\pi \sqrt{\frac{l}{g}}$ . Find the maximum(%) error in T due to the possible error up to 1% in  $l$  and 2.5 % in  $g$ .

**AKTU 2014**

$$\frac{\delta l}{l} \times 100 = +1\%$$

$$\frac{\delta g}{g} \times 100 = -2.5\%$$

$$\frac{\delta T}{T} \times 100 = ?$$

$$T = 2\pi \sqrt{\frac{l}{g}}$$

Taking log both side

$$\log T = \log 2\pi \left(\frac{l}{g}\right)^{\frac{1}{2}}$$

$$\log T = \log 2\pi + \log \left(\frac{l}{g}\right)^{\frac{1}{2}}$$

$$\log T = \log 2\pi + \frac{1}{2} \log \left(\frac{l}{g}\right)$$

$$\log T = \log 2\pi + \frac{1}{2} [\log l - \log g]$$

$$\frac{1}{T} \delta T = 0 + \frac{1}{2} \left[ \frac{1}{l} \delta l - \frac{1}{g} \delta g \right]$$

$$\frac{\delta T}{T} \times 100 = \frac{1}{2} \left[ \frac{\delta l}{l} \times 100 - \frac{\delta g}{g} \times 100 \right]$$

$$\frac{\delta T}{T} \times 100 = \frac{1}{2} [1\% + 2.5\%]$$

$$\frac{\delta T}{T} = 1.75\%$$

**Q.5** What error in the common logarithm of a number will be produced by an error of 1% in the number ?

**AKTU 2018**

Let  $n$  be a number

$$\frac{\delta n}{n} \times 100 = 1\%$$

common logarithm of  $n$

$$u = \log_{10} n$$

$$\delta u = \frac{1}{n \log 10} \delta n$$

$$\delta u = \frac{\delta n}{n} \times \frac{1}{\log_{10} e}$$

$$\delta u = \frac{1}{100} \times \frac{1}{\log_{10} e}$$

$$\delta u = 0.0043429$$

$$\frac{d}{dn} (\log_a n) = \frac{1}{n \log a}$$

$$\log_y x = \frac{1}{\log_x y}$$



**Q.6** A balloon is in the form of right circular cylinder of radius 1.5 m and length 4 m and is surmounted by hemispherical ends. If the radius is increased by 0.01 m and length by 0.05 m, find the percentage change in the volume of balloon

**AKTU 2014, 2018**

Given

$$r = 1.5 \text{ m}$$

$$h = 4 \text{ m}$$

$$\delta r = 0.01 \text{ m}$$

$$\delta h = 0.05 \text{ m}$$

Volume of balloon

$$V = \pi r^2 h + \frac{2}{3} \pi r^3 + \frac{2}{3} \pi r^3$$

$$V = \pi r^2 h + \frac{4}{3} \pi r^3$$

By Total differentiation

$$\delta V = (2\pi r h + 4\pi r^2) \delta r + \pi r^2 \delta h$$

$$\delta V = \pi r (2h \delta r + 4r \delta r + r \delta h)$$



$$\frac{\delta V}{V} = \frac{\pi r (2h \delta r + 4r \delta r + r \delta h)}{V}$$

$$\frac{\delta V}{V} \times 100 = \frac{\pi r (2h \delta r + 4r \delta r + r \delta h)}{(\pi r^2 h + \frac{4}{3} \pi r^3)} \times 100$$

$$\frac{\delta V}{V} \times 100 = \frac{\cancel{\pi r} (2h \delta r + 4r \delta r + r \delta h)}{\cancel{\pi r} (rh + \frac{4}{3} r^2)}$$

$$\frac{\delta V}{V} \times 100 = \frac{2 \times 4 \times 0.01 + 4 \times 1.5 \times 0.01 + 1.5 \times 0.05}{1.5 \times 4 + \frac{4}{3} 1.5 \times 1.5} \times 100$$

$$\frac{\delta V}{V} \times 100 = 2.389 \%$$



**Q.7** Find the possible percentage error in computing the parallel resistance  $r$  of three

resistances  $r_1, r_2, r_3$  from the formula  $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$  if  $r_1, r_2, r_3$  are each in error by  $+1.2\%$

Given

$$\frac{\delta r_1}{r_1} \times 100 = 1.2\%$$

$$\frac{\delta r_2}{r_2} \times 100 = 1.2\%$$

$$\frac{\delta r_3}{r_3} \times 100 = 1.2\%$$

$$\frac{\delta r}{r} \times 100 = ?$$

$$\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$$

By Total Differentiation

$$-\frac{1}{r^2} \delta r = -\frac{1}{r_1^2} \delta r_1 - \frac{1}{r_2^2} \delta r_2 - \frac{1}{r_3^2} \delta r_3$$

$$\frac{\delta r}{r^2} = \frac{\delta r_1}{r_1^2} + \frac{\delta r_2}{r_2^2} + \frac{\delta r_3}{r_3^2}$$

$$\frac{1}{r} \left( \frac{\delta r}{r} \right) = \frac{1}{r_1} \left( \frac{\delta r_1}{r_1} \right) + \frac{1}{r_2} \left( \frac{\delta r_2}{r_2} \right) + \frac{1}{r_3} \left( \frac{\delta r_3}{r_3} \right)$$

$$\frac{1}{r} \left( \frac{\delta r}{r} \times 100 \right) = \frac{1}{r_1} \left( \frac{\delta r_1}{r_1} \times 100 \right) + \frac{1}{r_2} \left( \frac{\delta r_2}{r_2} \times 100 \right) + \frac{1}{r_3} \left( \frac{\delta r_3}{r_3} \times 100 \right)$$

$$\frac{1}{r} \left( \frac{\delta r}{r} \times 100 \right) = \frac{1}{r_1} \times 1.2 + \frac{1}{r_2} \times 1.2 + \frac{1}{r_3} \times 1.2$$

$$\frac{1}{r} \left( \frac{\delta r}{r} \times 100 \right) = \left( \frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} \right) 1.2\%$$

$$\frac{1}{r} \left( \frac{\delta r}{r} \times 100 \right) = \frac{1}{r} \times 1.2\%$$

$$\frac{\delta r}{r} \times 100 = 1.2\%$$

## Type-III : Approximate value of given function

### Step to solve question

**STEP – 1** : Consider the function

$f(x)$  for one variable

$f(x, y)$  for two variable

$f(x, y, z)$  for three variable

**STEP – 2** : Round off each term to its nearest integer and identify  $x$ ,  $dx$ ,  $y$ ,  $dy$ ,  $z$ ,  $dz$

**STEP – 3** : Put above value in  $du = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$

**STEP – 4** : Approximate value of given function is  $u + du$



**Q.8** Compute an approximate value of  $(1.04)^{3.01}$

**AKTU 2014**

Let  $f(x, y) = x^y$

on comparison

$$x = 1$$

$$dx = 0.04$$

$$y = 3$$

$$dy = 0.01$$

$$df = \left( \frac{\partial f}{\partial x} \right) dx + \frac{\partial f}{\partial y} dy$$

$$df = y x^{y-1} dx + x^y \log x dy$$

$$df = 3 \times (1)^2 \times 0.04 + (1)^3 \log 1 \times 0.01$$

$$df = 0.12$$

$$f(1, 3) = (1)^3 = 1$$

$$df + f = 0.12 + 1$$

$$(1.04)^{3.01} = 1.12$$

**Q.9** Find approximate value of  $[(0.98)^2 + (2.01)^2 + (1.94)^2]^{1/2}$

**AKTU 2015**

Let  $f(x, y, z) = (x^2 + y^2 + z^2)^{1/2}$

on comparison

$x = 1, dx = -0.02$

$y = 2, dy = 0.01$

$z = 2, dz = -0.06$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$df = \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2} \times 2x dx + \frac{1}{2}(x^2 + y^2 + z^2)^{-1/2} \times 2y dy +$$

$$\frac{1}{2}(x^2 + y^2 + z^2)^{-1/2} \times 2z dz$$

$$df = (x^2 + y^2 + z^2)^{-1/2} (x dx + y dy + z dz)$$

$$df = (1^2 + 2^2 + 2^2)^{-1/2} (-0.02 + 2 \times 0.01 - 2 \times 0.06)$$

$$df = \frac{1}{(9)^{1/2}} \times (-0.12)$$

$$df = -0.04$$



$$\|(1, 2, 2)\| = (1^2 + 2^2 + 2^2)^{1/2}$$

$$= (9)^{1/2}$$

$$= 3$$

$$\|+d\| = 3 - 0.04$$

$$[(0.98)^2 + (2.01)^2 + (1.94)^2]^{1/2} = 2.96$$

## UNIT -3 : DPP - 7

## Topic : Approximation of Error

Q.1 Find the percentage error in the area of a rectangle when an error of +1 percent is made in measuring its length and breadth.

Q.2 Find the percentage error in the area of an ellipse when an error of +1 % is made in measuring the major and minor axes.

$$A = \pi xy$$

Q.3 If the base radius and height of a cone are measured as 4 and 8 inches with a possible error of 0.04 and 0.08 inches respectively, calculate the percentage (%) error in calculating volume of the cone.

Q.4 In estimating the cost of a pile of bricks measured as 6 m x 50 m x 4 m, the tape is stretched 1% beyond the standard length. If the count is 12 bricks in 1 m<sup>3</sup> and bricks cost Rs.100 per 1,000, find the approximate error in the cost.



## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 8

#### Today's target

- JACOBIAN : INTRODUCTION
- PYQs
- DPP



By Gulshan sir

# JACOBIANS

Basics of Jacobian

L-8

Properties of Jacobian

L-9

Jacobian of Implicit functions

L-10

V. Imp

## JACOBIANS

Jacobian is a functional determinant obtain by partial differentiation of two or more dependent variables w. r. t. two or more independent variables



(i) For two variables

If  $u = f_1(x, y)$  and  $v = f_2(x, y)$  then Jacobian of  $u, v$  with respect to  $x, y$  is defined as

$$J(u, v) = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

If  $u = f_1(x, y, z)$  and  $v = f_2(x, y, z)$  and  $w = f_3(x, y, z)$  then Jacobian of  $u, v, w$  with respect to  $x, y, z$  is defined as

$$J(u, v, w) = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$



Q.1 If  $u = x(1 - y)$ ,  $v = xy$ , find  $\frac{\partial(u,v)}{\partial(x,y)}$

AKTU 2020

$$u = x(1 - y) = x - xy$$

$$v = xy$$

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 1 - y & -x \\ y & x \end{vmatrix}$$

$$= (1 - y)x + xy$$

$$\frac{\partial(u,v)}{\partial(x,y)} = x - xy + xy$$

$$\frac{\partial(u,v)}{\partial(x,y)} = x$$

Q.2 If  $x = e^v \sin u$ ,  $y = e^v \cos u$ , then find  $\frac{\partial(x, y)}{\partial(u, v)}$

$$x = e^v \sin u$$

$$y = e^v \cos u$$

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

$$= \begin{vmatrix} e^v \cos u & e^v \sin u \\ -e^v \sin u & e^v \cos u \end{vmatrix}$$

$$\frac{\partial(x, y)}{\partial(u, v)} = (e^v \cos u)^2 + (e^v \sin u)^2$$

$$= e^{2v} \cos^2 u + e^{2v} \sin^2 u$$

$$= e^{2v} (\cos^2 u + \sin^2 u)$$

$$\frac{\partial(x, y)}{\partial(u, v)} = e^{2v}$$



Q.3 Calculate the Jacobian  $\frac{\partial(u,v,w)}{\partial(x,y,z)}$  of following:  $u = x + 2y + z$ ,  $v = x + 2y + 3z$ ,

$$w = 2x + 3y + 5z$$

$$u = x + 2y + z$$

$$v = x + 2y + 3z$$

$$w = 2x + 3y + 5z$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 5 \end{vmatrix}$$

$$= 1(10 - 9) - 2(5 - 6) + 1(3 - 4)$$

$$= 1 + 2 - 1$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = 2$$

Q.4 If  $y_1 = \frac{x_2 x_3}{x_1}$ ,  $y_2 = \frac{x_1 x_3}{x_2}$ ,  $y_3 = \frac{x_1 x_2}{x_3}$ , then show that  $\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = 4$

AKTU-2018

$$y_1 = \frac{x_2 x_3}{x_1}$$

$$y_2 = \frac{x_1 x_3}{x_2}$$

$$y_3 = \frac{x_1 x_2}{x_3}$$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \\ \frac{\partial y_3}{\partial x_1} & \frac{\partial y_3}{\partial x_2} & \frac{\partial y_3}{\partial x_3} \end{vmatrix}$$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \begin{vmatrix} -\frac{x_2 x_3}{x_1^2} & \frac{x_3}{x_1} & \frac{x_2}{x_1} \\ \frac{x_3}{x_2} & -\frac{x_1 x_3}{x_2^2} & \frac{x_1}{x_2} \\ \frac{x_2}{x_3} & \frac{x_1}{x_3} & -\frac{x_1 x_2}{x_3^2} \end{vmatrix}$$

Taking  $\frac{1}{x_1^2}$ ,  $\frac{1}{x_2^2}$ ,  $\frac{1}{x_3^2}$  common from

$R_1, R_2, R_3$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \frac{1}{x_1^2 x_2^2 x_3^2} \begin{vmatrix} -x_2 x_3 & x_3 x_1 & x_2 x_1 \\ x_2 x_3 & -x_1 x_3 & x_2 x_1 \\ x_2 x_3 & x_1 x_3 & -x_1 x_2 \end{vmatrix}$$



Taking  $x_2 x_3$ ,  $x_3 x_1$  and  $x_2 x_1$  common from  $c_1$ ,  $c_2$  and  $c_3$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \frac{\cancel{x_2} \cancel{x_3} \times \cancel{x_3} \cancel{x_1} \times \cancel{x_2} \cancel{x_1}}{\cancel{x_1}^2 \times \cancel{x_2}^2 \times \cancel{x_3}^2} \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= -1(1-1) - 1(-1-1) + 1(1+1)$$

$$= 0 + 2 + 2$$

$$\boxed{\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = 4}$$

Hence proved

## UNIT -3 : DPP - 8

## Topic : JACOBIAN

Q.1 If  $u = e^v \sec u$ ,  $y = e^v \tan u$ , then evaluate  $\frac{\partial(x,y)}{\partial(u,v)}$

AKTU 2021

Q.2 If  $u = xyz$ ,  $v = xy + yz + zx$ ,  $w = x + y + z$ , then compute the Jacobian  $\frac{\partial(u,v,w)}{\partial(x,y,z)}$

Q.3 If  $x = r \cos \theta$ ,  $y = r \sin \theta$ ,  $z = z$ , find  $\frac{\partial(x,y,z)}{\partial(r,\theta,z)}$ .

AKTU-2019



## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 9

#### Today's target

- PROPERTIES OF JACOBIAN
- PYQs
- DPP



By Gulshan sir

(i) For two variables

If  $u = f_1(x, y)$  and  $v = f_2(x, y)$  then Jacobian of  $u, v$  with respect to  $x, y$  is defined as

$$J(u, v) = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$



If  $u = f_1(x, y, z)$  and  $v = f_2(x, y, z)$  and  $w = f_3(x, y, z)$  then Jacobian of  $u, v, w$  with respect to  $x, y, z$  is defined as

$$J(u, v, w) = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

## Important properties of determinant

(1) If any two rows (or columns) of a determinant are identical, then the value of determinant is Zero

(2) We can take common from any Row or Column

(3) We can apply the following operations

(i) Row operation

$$R_i \rightarrow R_i + kR_j$$

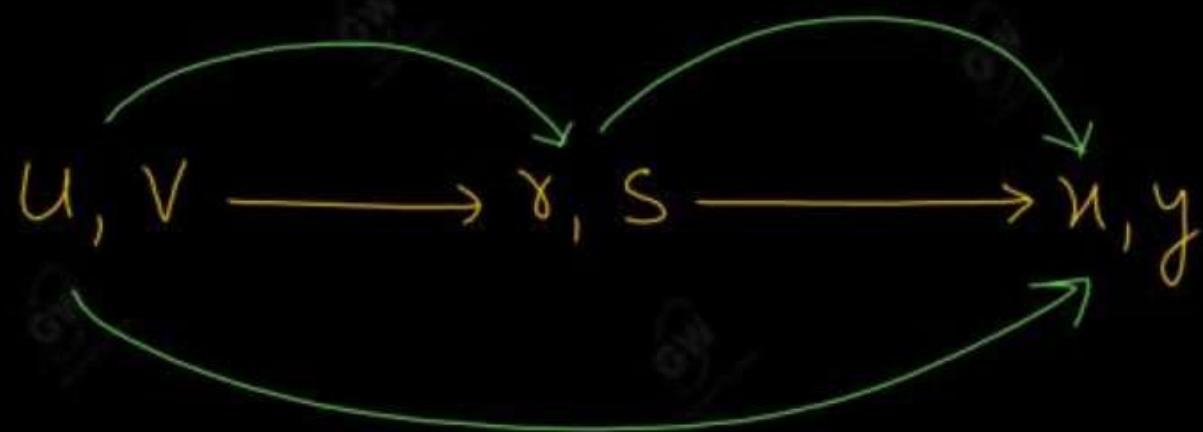
(ii) Column operation

$$C_i \rightarrow C_i + kC_j$$



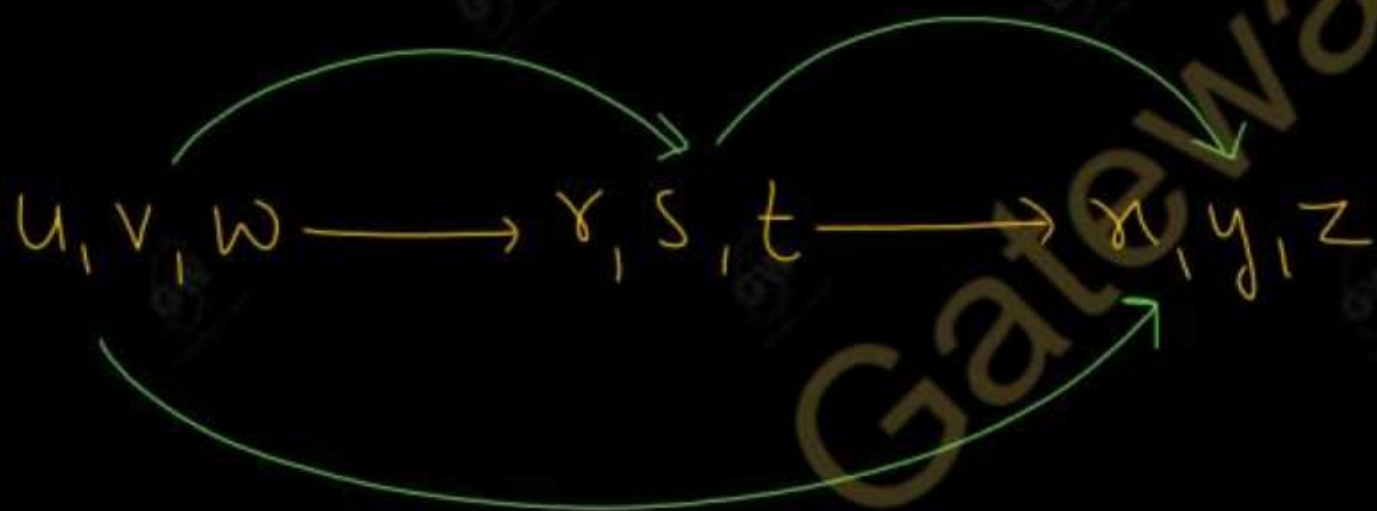
# (1) Chain Rule of Jacobian

(i) If  $u, v$  are the function of  $r, s$  and  $r, s$  are the function of  $x, y$



$$\frac{\partial(u, v)}{\partial(x, y)} = \frac{\partial(u, v)}{\partial(r, s)} \times \frac{\partial(r, s)}{\partial(x, y)}$$

(ii) If  $u, v, w$  are the function of  $r, s, t$  and  $r, s, t$  are the function of  $x, y, z$



$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{\partial(u, v, w)}{\partial(r, s, t)} \times \frac{\partial(r, s, t)}{\partial(x, y, z)}$$

**Q.1** If  $x = \sqrt{vw}$ ,  $y = \sqrt{wu}$ ,  $z = \sqrt{uv}$  and  $u = r \sin \theta \cos \phi$ ,  $v = r \sin \theta \sin \phi$ ,  $w = r \cos \theta$ , then calculate the Jacobian  $\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}$

**AKTU 2015**

$$x = \sqrt{vw}$$

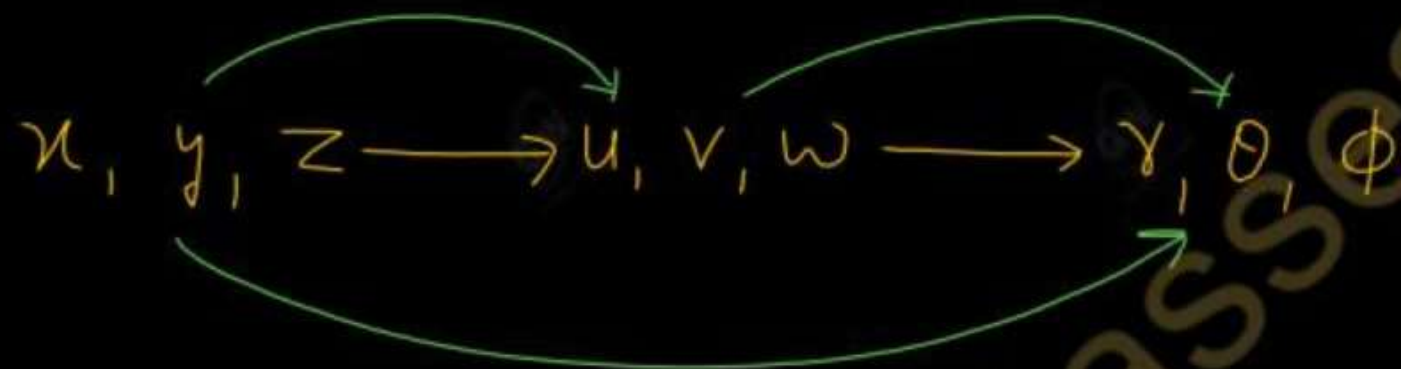
$$y = \sqrt{wu}$$

$$z = \sqrt{uv}$$

$$u = r \sin \theta \cos \phi$$

$$v = r \sin \theta \sin \phi$$

$$w = r \cos \theta$$



$$\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = \frac{\partial(x,y,z)}{\partial(u,v,w)} \times \frac{\partial(u,v,w)}{\partial(r,\theta,\phi)} \quad \text{--- (1)}$$

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix} = \begin{vmatrix} 0 & \frac{w}{2\sqrt{vw}} & \frac{v}{2\sqrt{vw}} \\ \frac{w}{2\sqrt{wu}} & 0 & \frac{u}{2\sqrt{wu}} \\ \frac{v}{2\sqrt{uv}} & \frac{u}{2\sqrt{uv}} & 0 \end{vmatrix}$$



Taking  $\frac{1}{2\sqrt{vw}}$ ,  $\frac{1}{2\sqrt{wu}}$ ,  $\frac{1}{2\sqrt{uv}}$  common

from  $R_1, R_2, R_3$

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \frac{1}{2\sqrt{vw} \times 2\sqrt{wu} \times 2\sqrt{uv}} \begin{vmatrix} 0 & w & v \\ w & 0 & u \\ v & u & 0 \end{vmatrix}$$

$$= \frac{1}{8\sqrt{vw} \times wu \times uv} [0 - w(0 - uv) + v(wu - 0)]$$

$$= \frac{1}{8\sqrt{u^2 v^2 w^2}} \times (uvw + uvw)$$

$$= \frac{1}{8uvw} \times 2uvw$$

4

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \frac{1}{4} \quad \text{--- (2)}$$

$$\frac{\partial(u, v, w)}{\partial(r, \theta, \phi)} = \begin{vmatrix} \frac{\partial u}{\partial r} & \frac{\partial u}{\partial \theta} & \frac{\partial u}{\partial \phi} \\ \frac{\partial v}{\partial r} & \frac{\partial v}{\partial \theta} & \frac{\partial v}{\partial \phi} \\ \frac{\partial w}{\partial r} & \frac{\partial w}{\partial \theta} & \frac{\partial w}{\partial \phi} \end{vmatrix}$$

$$= \begin{vmatrix} \sin \theta \cos \phi & r \cos \theta \cos \phi & -r \sin \theta \sin \phi \\ \sin \theta \sin \phi & r \cos \theta \sin \phi & r \sin \theta \cos \phi \\ \cos \theta & -r \sin \theta & 0 \end{vmatrix}$$

Taking  $r$  and  $r \sin \theta$  common  
from  $c_2$  and  $c_3$

$$\frac{\partial(u, v, w)}{\partial(r, \theta, \phi)} = r \times r \sin \theta \begin{vmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{vmatrix}$$

Expand along  $R_1$

$$= r^2 \sin \theta \left[ \sin \theta \cos \phi (0 + \sin \theta \cos \phi) - \cos \theta \cos \phi (0 - \cos \theta \cos \phi) \right. \\ \left. - \sin \phi (-\sin^2 \theta \sin \phi - \cos^2 \theta \sin \phi) \right]$$

$$\frac{\partial(u, v, w)}{\partial(r, \theta, \phi)} = r^2 \sin \theta \left[ \sin^2 \theta \cos^2 \phi + \cos^2 \theta \cos^2 \phi + \sin^2 \theta \sin^2 \phi + \cos^2 \theta \sin^2 \phi \right]$$

$$= r^2 \sin \theta \left[ \cos^2 \phi (\sin^2 \theta + \cos^2 \theta) + \sin^2 \phi (\sin^2 \theta + \cos^2 \theta) \right]$$

$$= r^2 \sin \theta (\cos^2 \phi + \sin^2 \phi)$$

$$= r^2 \sin \theta$$



$$\frac{\partial(u, v, w)}{\partial(r, \theta, \phi)} = r^2 \sin \theta \quad \text{--- (3)}$$

Using eq<sup>n</sup> (2) and (3) in (1)

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \phi)} = \frac{1}{r} r^2 \sin \theta$$

Gateway Classes

## (2) Cyclic Chain Rule of Jacobian ( $J J' = 1$ )

(i) If  $J$  is the Jacobian of  $u, v$  with respect to  $x, y$  and  $J'$  is the Jacobian of  $x, y$  with respect to  $u, v$

$$J J' = 1$$

$$\frac{\partial(u, v)}{\partial(x, y)} \times \frac{\partial(x, y)}{\partial(u, v)} = 1$$

(ii) If  $J$  is the Jacobian of  $u, v, w$  with respect to  $x, y, z$  and  $J'$  is the Jacobian of  $x, y, z$  with respect to  $u, v, w$

$$J J' = 1$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} \times \frac{\partial(x, y, z)}{\partial(u, v, w)} = 1$$



Q.2 If  $u = xyz, v = x^2 + y^2 + z^2, w = x + y + z$ , find the jacobian  $\frac{\partial(x,y,z)}{\partial(u,v,w)}$

$$u = xyz$$

$$v = x^2 + y^2 + z^2$$

$$w = x + y + z$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = \begin{vmatrix} yz & xz & xy \\ 2x & 2y & 2z \\ 1 & 1 & 1 \end{vmatrix}$$

$$C_1 \rightarrow C_1 - C_3 \text{ and } C_2 \rightarrow C_2 - C_3$$

$$\begin{aligned} \frac{\partial(u,v,w)}{\partial(x,y,z)} &= \begin{vmatrix} yz - xy & xz - xy & xy \\ 2x - 2z & 2y - 2z & 2z \\ 0 & 0 & 1 \end{vmatrix} \\ &= \begin{vmatrix} y(z-x) & -x(y-z) & xy \\ -2(z-x) & 2(y-z) & 2z \\ 0 & 0 & 1 \end{vmatrix} \end{aligned}$$

Taking  $(z-x)$  and  $(y-z)$  common  
from  $c_1$  and  $c_2$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = (z-x)(y-z) \begin{vmatrix} y & -x & xy \\ -2 & 2 & 2z \\ 0 & 0 & 1 \end{vmatrix}$$

Expand along  $R_3$

$$\begin{aligned} \frac{\partial(u, v, w)}{\partial(x, y, z)} &= (z-x)(y-z) [1(2y-2x)] \\ &= -2(x-y)(y-z)(z-x) \end{aligned}$$

Apply Chain Rule

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} \times \frac{\partial(x, y, z)}{\partial(u, v, w)} = 1$$

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \frac{1}{\frac{\partial(u, v, w)}{\partial(x, y, z)}}$$

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \frac{-1}{2(x-y)(y-z)(z-x)}$$



Q.3 If  $x = v^2 + w^2, y = w^2 + u^2, z = u^2 + v^2$ , then show that  $\frac{\partial(x,y,z)}{\partial(u,v,w)} \times \frac{\partial(u,v,w)}{\partial(x,y,z)} = 1$

$$x = v^2 + w^2 \quad \text{--- (1)}$$

$$y = w^2 + u^2 \quad \text{--- (2)}$$

$$z = u^2 + v^2 \quad \text{--- (3)}$$

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

$$\frac{\partial(x,y,z)}{\partial(u,v,w)} = \begin{vmatrix} 0 & 2v & 2w \\ 2u & 0 & 2w \\ 2u & 2v & 0 \end{vmatrix}$$

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Taking  $2u, 2v, 2w$  common from  $C_1, C_2$  and  $C_3$

$$= 2u \times 2v \times 2w \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$= 8uvw [0 - 1(0-1) + 1(1-0)]$$

$$= 16uvw$$

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = 16uvw$$

Now

Adding ① and ②

$$x+y = v^2+w^2+w^2+u^2$$

$$x+y = 2w^2+v^2+u^2$$

$$x+y = 2w^2+z \quad (\text{using ③})$$

$$2w^2 = x+y-z$$

$$w = \sqrt{\frac{x+y-z}{2}}$$

Adding ② and ③

$$y+z = w^2+u^2+u^2+v^2$$

$$y+z = 2u^2+w^2+v^2$$

$$y+z = 2u^2+x \quad (\text{using ①})$$

$$2u^2 = y+z-x$$

$$u = \sqrt{\frac{y+z-x}{2}}$$

Adding ① and ③

$$x+z = v^2+w^2+u^2+v^2$$

$$x+z = 2v^2+w^2+u^2$$

$$x+z = 2v^2+y$$

$$2v^2 = x+z-y$$

$$v = \sqrt{\frac{x+z-y}{2}}$$



$$\checkmark w = \frac{1}{\sqrt{2}} \sqrt{x+y-z}$$

$$\checkmark u = \frac{1}{\sqrt{2}} \sqrt{y+z-x}$$

$$\checkmark v = \frac{1}{\sqrt{2}} \sqrt{x+z-y}$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{1}{\sqrt{2}} \times \frac{-1}{2\sqrt{y+z-x}} & \frac{1}{\sqrt{2}} \times \frac{1}{2\sqrt{y+z-x}} & \frac{1}{\sqrt{2}} \times \frac{1}{2\sqrt{y+z-x}} \\ \frac{1}{\sqrt{2}} \times \frac{1}{2\sqrt{x+z-y}} & \frac{1}{\sqrt{2}} \times \frac{-1}{2\sqrt{x+z-y}} & \frac{1}{\sqrt{2}} \times \frac{1}{2\sqrt{x+z-y}} \\ \frac{1}{\sqrt{2}} \times \frac{1}{2\sqrt{x+y-z}} & \frac{1}{\sqrt{2}} \times \frac{1}{2\sqrt{x+y-z}} & \frac{1}{\sqrt{2}} \times \frac{-1}{2\sqrt{x+y-z}} \end{vmatrix}$$

Taking common from  $R_1, R_2$  and  $R_3$

$$\begin{aligned} \frac{\partial(u, v, w)}{\partial(x, y, z)} &= \frac{1}{2\sqrt{2}(\sqrt{y+z-x}) \underbrace{2\sqrt{2}(\sqrt{x+z-y})}_{R_2} \underbrace{2\sqrt{2}(\sqrt{x+y-z})}_{R_3}} \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} \\ &= \frac{1}{16\sqrt{2}(\sqrt{(y+z-x)(x+z-y)(x+y-z)})} \times [-1(1-1) - 1(-1-1) + 1(1+1)] \end{aligned}$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{4}{4 \cancel{16} \sqrt{2} \sqrt{(y+z-x)(x+z-y)(x+y-z)}}$$

Put  $x, y, z$  from eq<sup>n</sup> ①, ② and ③ we get

$$\begin{aligned} \frac{\partial(u, v, w)}{\partial(x, y, z)} &= \frac{1}{4 \sqrt{2} \sqrt{2u^2 \times 2v^2 \times 2w^2}} \\ &= \frac{1}{4 \sqrt{2} \times 2 \sqrt{2} uvw} \end{aligned}$$

$$\boxed{\frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{1}{16uvw}} \quad \text{--- ⑤}$$

Multiply ④ and ⑤

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} \times \frac{\partial(u, v, w)}{\partial(x, y, z)} = \cancel{16uvw} \times \frac{1}{\cancel{16uvw}}$$

$$\boxed{\frac{\partial(x, y, z)}{\partial(u, v, w)} \times \frac{\partial(u, v, w)}{\partial(x, y, z)} = 1}$$

Hence proved



UNIT -3 : DPP - 9

Topic : PROPERTIES OF JACOBIAN

Q.1 If  $u = x^2 - y^2, v = 2xy$  and  $x = r \cos \theta, y = r \sin \theta$ , show that  $\frac{\partial(u,v)}{\partial(r,\theta)} = 4r^3$

Q.2 If  $x = r \cos \theta, y = r \sin \theta$  find  $\frac{\partial(x,y)}{\partial(r,\theta)}$  and  $\frac{\partial(r,\theta)}{\partial(x,y)}$ . Also prove that  $JJ' = 1$

Q.3 Verify the chain rule for Jacobians if  $x = u, y = u \tan v, z = w$

Gateway Classes

## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 10

#### Today's target

- PROPERTIES OF JACOBIAN (PART – 2)
- PYQs
- DPP



By Gulshan sir



### (3) Functional dependence

(i) Two functions  $u = f_1(x, y)$  and  $v = f_2(x, y)$  are functionally dependent if and only if

$$\frac{\partial(u, v)}{\partial(x, y)} = 0$$

(ii) Three functions  $u = f_1(x, y, z)$  and  $v = f_2(x, y, z)$   $w = f_3(x, y, z)$  are functionally dependent if and only if

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$$

**Q.1** Find the Jacobian of the functions :  $y_1 = (x_1 - x_2)(x_2 + x_3)$ ,  $y_2 = (x_1 + x_2)(x_2 - x_3)$ ,  $y_3 = x_2(x_1 - x_3)$  hence show that the functions are not independent. Find the relation between them

$$y_1 = x_1 x_2 + x_1 x_3 - x_2^2 - x_2 x_3 \quad \text{--- (1)}$$

$$y_2 = x_1 x_2 - x_1 x_3 + x_2^2 - x_2 x_3 \quad \text{--- (2)}$$

$$y_3 = x_1 x_2 - x_2 x_3 \quad \text{--- (3)}$$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \\ \frac{\partial y_3}{\partial x_1} & \frac{\partial y_3}{\partial x_2} & \frac{\partial y_3}{\partial x_3} \end{vmatrix}$$

**AKTU 2023**

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \begin{vmatrix} x_2 + x_3 & x_1 - 2x_2 - x_3 & x_1 - x_2 \\ x_2 - x_3 & x_1 + 2x_2 - x_3 & -x_1 - x_2 \\ x_2 & x_1 - x_3 & -x_2 \end{vmatrix}$$

$$C_1 \rightarrow C_1 + C_3$$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \begin{vmatrix} x_1 + x_3 & x_1 - 2x_2 - x_3 & x_1 - x_2 \\ -(x_1 + x_3) & x_1 + 2x_2 - x_3 & -x_1 - x_2 \\ 0 & x_1 - x_3 & -x_2 \end{vmatrix}$$



Taking  $(n_1 + n_3)$  common from  $C_1$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(n_1, n_2, n_3)} = (n_1 + n_3) \begin{vmatrix} 1 & n_1 - 2n_2 - n_3 & n_1 - n_2 \\ -1 & n_1 + 2n_2 - n_3 & -n_1 - n_2 \\ 0 & n_1 - n_3 & -n_2 \end{vmatrix}$$

$$R_2 \rightarrow R_2 + R_1$$

$$= (n_1 + n_3) \begin{vmatrix} 1 & n_1 - 2n_2 - n_3 & n_1 - n_2 \\ 0 & 2(n_1 - n_3) & -2n_2 \\ 0 & n_1 - n_3 & -n_2 \end{vmatrix}$$

Expand along  $C_1$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(n_1, n_2, n_3)} = (n_1 + n_3) \left[ -2n_2(n_1 - n_3) + 2n_2(n_1 - n_3) \right]$$

$$= (n_1 + n_3) (-\cancel{2n_1n_2} + \cancel{2n_2n_3} + \cancel{2n_1n_2} - \cancel{2n_2n_3})$$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(n_1, y_1, z_1)} = 0$$

Hence  $y_1, y_2, y_3$  are functionally not independent

Relation between  $y_1, y_2$  and  $y_3$

Adding eq<sup>n</sup> ①, ② and ③

$$y_1 + y_2 + y_3 = 3x_1x_2 - 3(x_2x_3)$$

$$y_1 + y_2 + y_3 = 3x_2(x_1 - x_3)$$

$$y_1 + y_2 + y_3 = 3y_3$$

$$y_1 + y_2 - 2y_3 = 0$$



**Q.5** Show that the functions:  $u = x + y + z$ ,  $v = x^2 + y^2 + z^2 - 2xy - 2yz - 2zx$  and,  $w = x^3 + y^3 + z^3 - 3xyz$  are functionally related. Find the relation between them

$$u = x + y + z \quad \text{--- (1)}$$

$$v = x^2 + y^2 + z^2 - 2xy - 2yz - 2zx \quad \text{--- (2)}$$

$$w = x^3 + y^3 + z^3 - 3xyz \quad \text{--- (3)}$$

$u, v, w$  are functionally related if  
and only if

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$$

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$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 2x-2y-2z & 2y-2x-2z & 2z-2y-2x \\ 3x^2-3yz & 3y^2-3xz & 3z^2-3xy \end{vmatrix}$$

$$C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} 1 & 0 & 0 \\ 2x-2y-2z & -4x+4y & -4x+4z \\ 3x^2-3yz & 3y^2-3x^2-3xz+3yz & 3z^2-3x^2-3xy+3yz \end{vmatrix}$$

Expand along  $R_1$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = 1 \left[ -4(x-y) 3(z^2-x^2-xy+yz) + 4(x-z) 3(y^2-x^2-xz+yz) \right]$$

$$= -12 \left[ \cancel{xz^2} - \cancel{x^3} - \cancel{xy} + \cancel{xyz} - \cancel{yz^2} + \cancel{yx^2} + \cancel{xy^2} - \cancel{yz^2} - \cancel{xy^2} + \cancel{x^3} + \cancel{x^2z} - \cancel{xyz} + \cancel{zy^2} - \cancel{zx^2} - \cancel{yz^2} + \cancel{yz^2} \right]$$



$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = 0$$

Hence

$u, v, w$  are functionally related

Relation between  $u, v, w$

Using eqn ①, ② and ③

$$w = x^3 + y^3 + z^3 - 3xyz$$

$$w = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$$

$$w = u(x^2+y^2+z^2-xy-yz-zx)$$

$$4w = u(4x^2+4y^2+4z^2-4xy-4yz-4zx)$$

$$4w = u(\underbrace{x^2+y^2+z^2-2xy-2yz-2zx}_{3x^2+3y^2+3z^2-2xy-2yz-2zx} + 3x^2+3y^2+3z^2-2xy-2yz-2zx)$$

$$4w = u(v + \underbrace{x^2+y^2+z^2-2xy-2yz-2zx}_{2x^2+2y^2+2z^2} + 2x^2+2y^2+2z^2)$$

$$4w = u(v + v + 2x^2+2y^2+2z^2)$$

$$4w = u(2v + 2x^2+2y^2+2z^2 + 2xy + 2yz + 2zx - 2xy - 2yz - 2zx)$$

$$4w = u(2v + \underbrace{x^2 + y^2 + z^2 - 2xy - 2yz - 2zx} + \underbrace{x^2 + y^2 + z^2 + 2xy + 2yz + 2zx})$$

$$4w = u[2v + v + (x+y+z)^2]$$

$$4w = u(3v + u^2)$$

$$w = \frac{u(3v + u^2)}{4}$$

Required Relation



## (4) When one variable increases in the successive function

$$u_1 = f_1(x_1)$$

$$u_2 = f_2(x_1, x_2)$$

$$u_3 = f_3(x_1, x_2, x_3)$$

|  
|  
|  
|  
|

$$u_n = f_n(x_1, x_2, x_3 \dots \dots \dots x_n)$$

$$\frac{\partial(u_1, u_2, u_3, \dots, u_n)}{\partial(x_1, x_2, x_3, \dots, x_n)} = \frac{\partial u_1}{\partial x_1} \times \frac{\partial u_2}{\partial x_2} \times \frac{\partial u_3}{\partial x_3} \dots \dots \frac{\partial u_n}{\partial x_n}$$

Q.6 If  $y_1 = \cos x_1$ ,  $y_2 = \sin x_1 \cos x_2$  and  $y_3 = \sin x_1 \sin x_2 \cos x_3$ , then show that

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = -\sin^3 x_1 \sin^2 x_2 \sin x_3$$

$$y_1 = \cos x_1$$

$$y_2 = \sin x_1 \cos x_2$$

$$y_3 = \sin x_1 \sin x_2 \cos x_3$$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = \frac{\partial y_1}{\partial x_1} \times \frac{\partial y_2}{\partial x_2} \times \frac{\partial y_3}{\partial x_3}$$

$$= (-\sin x_1) (-\sin x_1 \sin x_2) (-\sin x_1 \sin x_2 \sin x_3)$$

$$\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)} = -\sin^3 x_1 \sin^2 x_2 \sin x_3$$

Hence proved



## UNIT -3 : DPP - 10

## Topic : PROPERTIES OF JACOBIAN

**Q.1** If  $u = \frac{x+y}{1-xy}$  and  $v = \tan^{-1}x + \tan^{-1}y$ , find  $\frac{\partial(u,v)}{\partial(x,y)}$ . Are  $u$  and  $v$  functionally related? If so, find the relationship

**Q.2** If  $u = x + 2y + z$ ,  $v = x - 2y + 3z$  and  $w = 2xy - xz + 4yz - 2z^2$ , Show that they are not independent. Find the relation between  $u$ ,  $v$  and  $w$ .

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**Q.3** If  $y_1 = 1 - x_1$ ,  $y_2 = x_1(1 - x_2)$ ,  $y_3 = x_1x_2(1 - x_3)$ , find the value of  $\frac{\partial(y_1, y_2, y_3)}{\partial(x_1, x_2, x_3)}$ .

## ENGINEERING MATHEMATICS - I

### UNIT -3 : Differential calculus - II

#### Lecture - 11

#### Today's target

- JACOBIAN OF IMPLICIT FUNCTION
- PYQs
- DPP



By Gulshan sir



The function in which dependent and independent variable can not be separated in L.H.S and R.H.S

(i) If  $u$  and  $v$  are the implicit function of  $x$  and  $y$  then

$F_1 \equiv$  First equation

$F_2 \equiv$  Second equation

$$\frac{\partial(u, v)}{\partial(x, y)} = (-1)^2 \frac{\left[ \frac{\partial(F_1, F_2)}{\partial(x, y)} \right]}{\left[ \frac{\partial(F_1, F_2)}{\partial(u, v)} \right]}$$

(ii) If  $u, v$  and  $w$  are the implicit function of  $x, y$  and  $z$  then

$F_1 \equiv$  First equation

$F_2 \equiv$  Second equation

$F_3 \equiv$  Third equation

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = (-1)^3 \frac{\left[ \frac{\partial(F_1, F_2, F_3)}{\partial(x, y, z)} \right]}{\left[ \frac{\partial(F_1, F_2, F_3)}{\partial(u, v, w)} \right]}$$



Q.1 If  $u^3 + v^3 = x + y$ ,  $u^2 + v^2 = x^3 + y^3$ , then show that  $\frac{\partial(u, v)}{\partial(x, y)} = \frac{y^2 - x^2}{2uv(u - v)}$

$$u^3 + v^3 = x + y$$

$$u^2 + v^2 = x^3 + y^3$$

Here  $u, v$  are implicit function of  $x, y$

Let

$$F_1 \equiv u^3 + v^3 - x - y$$

$$F_2 \equiv u^2 + v^2 - x^3 - y^3$$

$$\frac{\partial(u, v)}{\partial(x, y)} = (-1)^2 \frac{\left[ \frac{\partial(F_1, F_2)}{\partial(x, y)} \right]}{\left[ \frac{\partial(F_1, F_2)}{\partial(u, v)} \right]} \quad \text{--- (1)}$$

$$\begin{aligned} \frac{\partial(F_1, F_2)}{\partial(x, y)} &= \begin{vmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{vmatrix} \\ &= \begin{vmatrix} -1 & -1 \\ -3x^2 & -3y^2 \end{vmatrix} \end{aligned}$$

AKTU 2024

$$\frac{\partial(F_1, F_2)}{\partial(x, y)} = 3y^2 - 3x^2$$

$$\frac{\partial(F_1, F_2)}{\partial(u, v)} = 3(y^2 - x^2) \quad \text{--- (2)}$$

$$\frac{\partial(F_1, F_2)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{vmatrix}$$

$$\frac{\partial(F_1, F_2)}{\partial(u, v)} = \begin{vmatrix} 3u^2 & 3v^2 \\ 2u & 2v \end{vmatrix}$$

$$= 6u^2v - 6uv^2$$

$$\boxed{\frac{\partial(F_1, F_2)}{\partial(u, v)} = 6uv(u-v)}$$

L (3)

Put eq<sup>n</sup> ② and ③ in ①

$$\frac{\partial(u, v)}{\partial(x, y)} = (-1)^2 \frac{\cancel{3}(y^2 - x^2)}{2uv(u-v)}$$

$$\boxed{\frac{\partial(u, v)}{\partial(x, y)} = \frac{y^2 - x^2}{2uv(u-v)}}$$

Hence proved



Q.2 If  $u, v, w$  are the roots of the cubic  $(\lambda - x)^3 + (\lambda - y)^3 + (\lambda - z)^3 = 0$  in  $\lambda$ , then

find  $\frac{\partial(u, v, w)}{\partial(x, y, z)}$

$$(a-b)^3 = a^3 - b^3 - 3a^2b + 3ab^2$$

AKTU 2016, 2022

$$(\lambda - x)^3 + (\lambda - y)^3 + (\lambda - z)^3 = 0$$

$$\lambda^3 - x^3 - 3\lambda^2x + 3\lambda x^2 + \lambda^3 - y^3 - 3\lambda^2y + 3\lambda y^2 + \lambda^3 - z^3 - 3\lambda^2z + 3\lambda z^2 = 0$$

$$3\lambda^3 - 3\lambda^2x - 3\lambda^2y - 3\lambda^2z + 3\lambda x^2 + 3\lambda y^2 + 3\lambda z^2 - x^3 - y^3 - z^3 = 0$$

$$3\lambda^3 - 3(x+y+z)\lambda^2 + 3(x^2+y^2+z^2)\lambda - (x^3+y^3+z^3) = 0$$

Here

$$a = 3$$

$$b = -3(x+y+z)$$

$$c = 3(x^2+y^2+z^2)$$

$$d = -(x^3+y^3+z^3)$$

Note

If  $\alpha, \beta, \gamma$  are the roots of  $ax^3 + bx^2 + cx + d = 0$  then

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$u+v+w = \frac{-b}{a} = \frac{-(x+y+z)}{3}$$

$$u+v+w = x+y+z \quad \text{--- (1)}$$

$$uv+vw+wu = \frac{c}{a} = \frac{-(x^2+y^2+z^2)}{3}$$

$$uv+vw+wu = x^2+y^2+z^2 \quad \text{--- (2)}$$

$$uvw = -\frac{d}{a} = -\frac{(x^3+y^3+z^3)}{3}$$

$$uvw = \frac{1}{3}(x^3+y^3+z^3) \quad \text{--- (3)}$$



Let

$$F_1 = u + v + w - x - y - z$$

$$F_2 = uv + vw + wu - x^2 - y^2 - z^2$$

$$F_3 = uvw - \frac{1}{3}(x^3 + y^3 + z^3)$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = (-1)^3 \frac{\left[ \frac{\partial(F_1, F_2, F_3)}{\partial(x, y, z)} \right]}{\left[ \frac{\partial(F_1, F_2, F_3)}{\partial(u, v, w)} \right]}$$

④

$$\frac{\partial(F_1, F_2, F_3)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z} \\ \frac{\partial F_3}{\partial x} & \frac{\partial F_3}{\partial y} & \frac{\partial F_3}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -1 & -1 \\ -2x & -2y & -2z \\ -x^2 & -y^2 & -z^2 \end{vmatrix}$$

$C_2 \rightarrow C_2 - C_1$  and  $C_3 \rightarrow C_3 - C_1$

$$\frac{\partial(F_1, F_2, F_3)}{\partial(x, y, z)} = \begin{vmatrix} -1 & 0 & 0 \\ -2x & 2(x-y) & 2(x-z) \\ -x^2 & (x-y)(x+y) & (x-z)(x+z) \end{vmatrix}$$

Taking  $(n-y)$  and  $(n-z)$  common from  $c_2$  and  $c_3$

$$\frac{\partial(F_1, F_2, F_3)}{\partial(n, y, z)} = (n-y)(n-z) \begin{vmatrix} -1 & 0 & 0 \\ -2n & 2 & 2 \\ -n^2 & n+y & n+z \end{vmatrix}$$

$$= (n-y)(n-z) \left[ -1 \{ 2(n+z) - 2(n+y) \} \right]$$

$$= (n-y)(n-z) (-2n - 2z + 2n + 2y)$$

$$= (n-y)(n-z) 2(y-z)$$

$$\frac{\partial(F_1, F_2, F_3)}{\partial(n, y, z)} = -2(n-y)(y-z)(z-n) \quad \text{--- (5)}$$

$$\frac{\partial(F_1, F_2, F_3)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} & \frac{\partial F_1}{\partial w} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} & \frac{\partial F_2}{\partial w} \\ \frac{\partial F_3}{\partial u} & \frac{\partial F_3}{\partial v} & \frac{\partial F_3}{\partial w} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \textcircled{1} & \textcircled{1} \\ v+w & u+w & u+v \\ vw & uw & uv \end{vmatrix}$$



$$C_2 \rightarrow C_2 - C_1 \text{ and } C_3 \rightarrow C_3 - C_1$$

$$\frac{\Delta(F_1, F_2, F_3)}{\Delta(u, v, w)} = \begin{vmatrix} 1 & 0 & 0 \\ v+w & u-v & u-w \\ vw & w(u-v) & v(u-w) \end{vmatrix}$$

Taking  $(u-v)$  and  $(u-w)$  common from  $C_2$  and  $C_3$

$$= (u-v)(u-w) \begin{vmatrix} 1 & 0 & 0 \\ v+w & 1 & 1 \\ vw & w & v \end{vmatrix}$$

$$= (u-v)(u-w) [(v-w)]$$

$$\frac{\Delta(F_1, F_2, F_3)}{\Delta(u, v, w)} = -(u-v)(v-w)(w-u) \quad \text{--- (6)}$$

Put eq<sup>n</sup> (5) and (6) in (4)

$$\frac{\Delta(u, v, w)}{\Delta(x, y, z)} = \frac{(-1)^3 (-2)(x-y)(y-z)(z-x)}{-(u-v)(v-w)(w-u)}$$

$$\frac{\Delta(u, v, w)}{\Delta(x, y, z)} = \frac{-2(x-y)(y-z)(z-x)}{(u-v)(v-w)(w-u)}$$

AKTU 2021

Show that  $\frac{\partial(u,v,w)}{\partial(x,y,z)} = \frac{1-4(xy+yz+zx)+16xyz}{2-3(u^2+v^2+w^2)+27u^2v^2w^2}$

Let

$$F_1 = u^3 + v + w - x - y^2 - z^2$$

$$F_2 = u + v^3 + w - x^2 - y - z^2$$

$$F_3 = u + v + w^3 - x^2 - y^2 - z$$

$$\frac{\partial(u,v,w)}{\partial(x,y,z)} = (-1)^3 \frac{\left[ \frac{\partial(F_1, F_2, F_3)}{\partial(x, y, z)} \right]}{\left[ \frac{\partial(F_1, F_2, F_3)}{\partial(u, v, w)} \right]}$$

①

$$\frac{\partial(F_1, F_2, F_3)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z} \\ \frac{\partial F_3}{\partial x} & \frac{\partial F_3}{\partial y} & \frac{\partial F_3}{\partial z} \end{vmatrix}$$

$$= \begin{vmatrix} -1 & -2y & -2z \\ -2x & -1 & -2z \\ -2x & -2y & -1 \end{vmatrix}$$

$$= -1(1 - 4yz) + 2y(2x - 4xz) - 2z(4xy - 2x)$$



$$\frac{\partial(F_1, F_2, F_3)}{\partial(x, y, z)} = -1 + 4yz + 4ny - 8nyz - 8nyz + 4zx$$

$$\frac{\partial(F_1, F_2, F_3)}{\partial(x, y, z)} = -1 + 4(ny + yz + zn) - 16nyz \quad \text{--- (2)}$$

$$\frac{\partial(F_1, F_2, F_3)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} & \frac{\partial F_1}{\partial w} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} & \frac{\partial F_2}{\partial w} \\ \frac{\partial F_3}{\partial u} & \frac{\partial F_3}{\partial v} & \frac{\partial F_3}{\partial w} \end{vmatrix}$$

$$= \begin{vmatrix} 3u^2 & 1 & 1 \\ 1 & 3v^2 & 1 \\ 1 & 1 & 3w^2 \end{vmatrix}$$

$$\frac{\partial(F_1, F_2, F_3)}{\partial(u, v, w)} = 3u^2(9v^2w^2 - 1) - 1(3w^2 - 1) + (1 - 3v^2)$$

$$= 27u^2v^2w^2 - 3u^2 - 3w^2 + 1 + 1 - 3v^2$$

$$\frac{\partial(F_1, F_2, F_3)}{\partial(u, v, w)} = 2 - 3(u^2 + v^2 + w^2) + 27u^2v^2w^2$$

$$27u^2v^2w^2$$

$$\text{--- (3)}$$

Put eq<sup>n</sup> ② and ③ in ①

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{(-1)^3 [-1 + 4(xy + yz + zx) - 16xyz]}{2 - 3(u^2 + v^2 + w^2) + 27u^2v^2w^2}$$

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = \frac{1 - 4(xy + yz + zx) + 16xyz}{2 - 3(u^2 + v^2 + w^2) + 27u^2v^2w^2}$$

Hence proved



UNIT -3 : DPP - 11

Topic : JACOBIAN OF IMPLICIT FUNCTION

Q.1 If  $u^3 + v^3 + w^3 = x + y + z$ ,  $u^2 + v^2 + w^2 = x^3 + y^3 + z^3$  and  $u + v + w = x^2 + y^2 + z^2$ , Then

show that  $\frac{\partial(u,v,w)}{\partial(x,y,z)} = \frac{(x-y)(y-z)(z-x)}{(u-v)(v-w)(w-u)}$

AKTU 2021

Q.2 If  $u, v, w$  are the roots of the equation  $(x-a)^3 + (x-b)^3 + (x-c)^3 = 0$ , then find  $\frac{\partial(u,v,w)}{\partial(x,y,z)}$

AKTU 2015, 2019

Q.3 If  $x + y + z = u$ ,  $y + z = uv$ ,  $z = uvw$ , then show that  $\frac{\partial(x,y,z)}{\partial(u,v,w)} = u^2v$

AKTU-2016

$$\begin{array}{c|c|c} x+uv=u & y+uvw=uv & z=uvw \\ \hline x=u-uv & y=uv-uvw & \end{array}$$

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**Thank You**